PRODUCT REQUIREMENTS SPECIFICATION

Enterprise Knowledge Platform

Unified search, conversational answers, semantic enrichment and governed knowledge lifecycle for the New York Power Authority and the New York State Canal Corporation

Solicitation	Knowledge Management RFP Q26-7821RJM, Attachment 1, Section 2.1
Product	KMS Enterprise Knowledge Platform
Document version	1.0
Date	October 2026
Classification	Proposal document. Confidential to the NYPA evaluation team.

Contents

1 Introduction 3

1.1 Purpose 3

1.2 Scope 3

2 Functional Requirements 4

2.1 Cross-cutting requirements 4

2.2 Unified access and landing experience 5

2.3 Search and navigation 6

2.4 Repository connectors and permission synchronization 10

2.5 Personalization and role-aware experience 14

2.6 Conversational query and answering 18

2.7 Knowledge content coverage 21

2.8 Content enrichment and semantic layer integration 24

2.9 Content ingestion 28

2.10 Automated metadata generation 29

2.11 Governance and content lifecycle management 31

2.12 Analytics and dashboards 37

2.13 Integration and authoring 40

2.14 Administration and configuration 42

3 Non-Functional Requirements 44

3.1 Performance and scalability 44

3.2 Availability and disaster recovery 47

3.3 Security 48

3.4 Deployment and environments 53

3.5 Usability and accessibility 54

3.6 Operations and support 56

3.7 Data management and privacy 56

3.8 Responsible AI 58

4 Data, Integration and Analytics Design Requirements 60

Appendix A. User Roles 63

Appendix B. Requirements Traceability Matrix 64

Section 2.1 Enterprise KM Platform Capabilities 64

Section 2.4 Data, Integration and Analytics (Enterprise Knowledge Platform) 66

Section 6 Application and Security Architecture Guidelines 66

1 Introduction

1.1 Purpose

This Product Requirements Specification defines the functional and non-functional requirements of the KMS Enterprise Knowledge Platform (the “Platform”). It responds to the Enterprise Knowledge Platform capability described in Section 2.1 of Attachment 1 to NYPA RFP Q26-7821RJM. NYPA’s capabilities are broken down into specific product requirements. Each requirement has a description of its intent and the acceptance criteria used to confirm it works.

1.2 Scope

In scope

- All capabilities in RFP Attachment 1, Section 2.1 “Enterprise KM Platform Capabilities”: unified access, search and navigation; personalization and role-aware experience; conversational query and answering; knowledge content coverage; content enrichment and semantic layer integration; content ingestion; automated metadata generation; governance and content lifecycle management; analytics and dashboards; and integration and authoring.
- The taxonomy, ontology and knowledge graph capabilities from Section 2.2 that the Platform uses for search, navigation and answers.
- The Application Architecture and Security Architecture guidelines in Section 6 of Attachment 1, expressed as non-functional requirements.
- The Enterprise Knowledge Platform data, integration and analytics design items listed in Section 2.4 of Attachment 1.

Out of scope

- The Lessons Learned Management Tool (Section 2.3) as a standalone product. The Platform can index and surface lessons learned content as one of its knowledge sources.
- Knowledge asset creation from knowledge transfer events (video segmentation, extraction of action items and auto-drafting from recordings), described in Section 2.2.
- Commercial terms, staffing and project schedule.

2 Functional Requirements

This section specifies what the Platform must do. Requirements are grouped by the capability areas of RFP Attachment 1, Section 2.1. Each requirement gives its priority and the RFP clause it addresses, a description of its intent, the detailed requirements, and the acceptance criteria used to confirm it. “Mandatory” requirements are required for the initial production release. “Target capability” requirements are those the RFP labels as target capabilities. The user roles referred to are defined in Appendix A.

2.1 Cross-cutting requirements

GEN-01 Permission enforcement across all capabilities

Priority	Mandatory
RFP reference	2.1 Unified Access, Search and Navigation, item 2 (“respects original permissions”); Conversational Query and Answering, item 2

Description. Every capability in the Platform shares one obligation: a user must only ever encounter knowledge they are entitled to see. This applies to everything the Platform shows a user, including search results, answers, recommendations, notifications, previews and graph navigation, not only to search. NYPA content includes operationally sensitive material about generation and transmission assets, personnel matters and regulated information. One leaked title or AI-generated sentence built from restricted content would undermine trust in the whole Platform. This requirement makes permission enforcement a cross-cutting rule that every other requirement inherits.

Detailed requirements

- **GEN-01.1** A user must never see, be recommended, be notified about, be shown in analytics drill-downs, or receive in an AI-generated answer or citation any content, metadata, title, snippet or thumbnail that the user is not authorized to access in the Platform and in the originating source system at the time of the request.
- **GEN-01.2** Permission enforcement must be applied before ranking, before AI generation and before display, and not only as a filter applied afterwards.
- **GEN-01.3** Where the Platform cannot determine whether a user may access an item, it must treat the item as inaccessible.
- **GEN-01.4** The same enforcement must apply to every Platform API, including access by other NYPA applications on behalf of a user.

Acceptance criteria

1. A permission test matrix built with NYPA, covering direct grants, group grants, nested groups, sharing links, broken inheritance and revoked access, passes across search, answers, recommendations, facets, notifications, previews, graph navigation and APIs with zero unauthorized disclosures.

2.2 Unified access and landing experience

ACC-01 Customizable landing page

Priority	Mandatory
RFP reference	2.1 Unified Access, Search and Navigation, item 1

Description. The landing page is the front door to enterprise knowledge and the first thing every NYPA and Canals employee sees after signing in. It brings together search, the conversational assistant, announcements, featured content and personal items in one place, so employees no longer need to know which repository to visit first. NYPA and Canals have different audiences, operating contexts and brand needs. Administrators must therefore be able to build and target different versions of the page without development work, and change them as priorities shift.

Detailed requirements

- **ACC-01.1** The Platform must provide a landing page reachable from a single, stable URL and presented immediately after sign-in.
- **ACC-01.2** Administrators must be able to compose the landing page from configurable components without code, including: search and question box, announcements, featured content, topic hub tiles, quick links, recently updated content, trending content, personalized recommendations, the user's tasks, and embedded dashboards.
- **ACC-01.3** Administrators must be able to create multiple landing page variants (for example, NYPA corporate, Canals, a specific generation facility) and assign each variant to directory groups, business units or departments.
- **ACC-01.4** The Platform must support NYPA branding, including logo, color palette, typography and footer content, to meet the NYPA standard look and feel.
- **ACC-01.5** Administrators should be able to schedule announcements and featured content with start and end dates.
- **ACC-01.6** Landing page changes must be previewable before publication and must be versioned so that a previous layout can be restored.
- **ACC-01.7** All landing page components must comply with GEN-01; a component that has no permitted content for the user must be hidden or show an empty state rather than an error.
- **ACC-01.8** The landing page must render correctly in Chrome and Edge on Windows desktops and on iOS mobile devices.

Acceptance criteria

1. An administrator creates a Canals landing page variant, assigns it to a Canals directory group, and a Canals user sees that variant after sign-in while a corporate user sees the corporate variant.
2. An administrator adds, reorders and removes components and changes branding without code, previews the change, publishes it, and restores the prior version.
3. A user without access to a featured document does not see it in the featured content component.

ACC-02 Unified navigation and access

Priority	Mandatory
RFP reference	2.1 Unified Access, Search and Navigation

Description. Beyond the landing page, users need a consistent way to move between searching, asking, browsing, managing their saved items and acting on tasks. Users must also be able to reach knowledge from the tools they already work in, such as Microsoft Teams and SharePoint, and share links to it safely. This requirement defines the navigation framework and the rules for links into and out of the Platform. Moving between the Platform and source systems must never bypass the permissions of those source systems.

Detailed requirements

- **ACC-02.1** The Platform must provide a consistent global navigation across all pages, including search, ask, browse by topic, my knowledge (saved searches, watchlists, followed topics), my tasks, contribute, and analytics (for permitted roles).
- **ACC-02.2** Opening a search result or citation must take the user to the item in the Platform's preview with a direct link to open it in the source system, preserving source-system permissions.
- **ACC-02.3** The Platform should be available as a Microsoft Teams personal app and as a SharePoint web part so that users can search and ask in the flow of work.
- **ACC-02.4** The Platform must support deep links to searches, topic hubs and documents that can be shared in email and Teams; recipients without access see an access-denied message without content disclosure.

Acceptance criteria

1. A user navigates between search, topic hubs, saved items and tasks from any page.
2. A user opens a SharePoint result in preview and then in SharePoint with one click.
3. A shared deep link opened by an unauthorized user discloses no title, snippet or metadata.

2.3 Search and navigation

SRCH-01 Unified search across repositories

Priority	Mandatory for SharePoint; target capability for Content Server and ServiceNow
RFP reference	2.1 Unified Access, Search and Navigation, item 2

Description. Unified search lets an employee enter one query and find relevant knowledge wherever it lives: SharePoint Online, OpenText Content Server, ServiceNow knowledge bases or the intranet. Results from all sources are combined into one ranked list with consistent presentation, so users can judge relevance without

knowing where each item came from. Each user sees only what they can open in the original repository. SharePoint is mandatory for the initial release. Content Server and ServiceNow are target capabilities delivered through the same framework.

Detailed requirements

- **SRCH-01.1** A single query must search SharePoint Online sites, libraries, lists, pages and OneDrive locations designated by NYPA.
- **SRCH-01.2** A single query must search OpenText Content Server and ServiceNow knowledge bases when those connectors are enabled.
- **SRCH-01.3** Results from all sources must be merged into one ranked list using a common relevance model, not presented as separate per-source lists.
- **SRCH-01.4** Each result must show title, source system, location path, content type, last modified date, owner, a highlighted snippet, key taxonomy tags and a thumbnail where available.
- **SRCH-01.5** Results must be security-trimmed against the requesting user's effective permissions as synchronized from each source system (see CONN-02).
- **SRCH-01.6** Users must be able to restrict a search to one or more sources.
- **SRCH-01.7** The Platform must support keyword operators: exact phrase, AND, OR, NOT, prefix wildcard and field-scoped search (for example, owner:, type:, site:).
- **SRCH-01.8** The Platform must provide spelling correction, query suggestions (type-ahead) and "did you mean" alternatives, drawing suggestions only from content and terms the user can access.
- **SRCH-01.9** Results must support sorting by relevance, date modified and title, and pagination or infinite scroll.
- **SRCH-01.10** Superseded, archived and retired content must be excluded from default results and included only when the user explicitly selects to include it.

Acceptance criteria

1. A query for a term present in SharePoint, Content Server and ServiceNow test content returns results from all three sources in one ranked list.
2. Two users with different permissions issue the same query; each sees only the items they can open in the source system, verified against a permission test matrix of at least 50 items.
3. Field-scoped and Boolean queries return the expected result sets on a reference corpus.
4. Search response time meets NFR-PERF-01.

SRCH-02 Semantic and hybrid search

Priority	Mandatory
RFP reference	2.1 Unified Access, Search and Navigation, item 3

Description. Employees often describe what they need in words that differ from the words in the documents, use acronyms that are specific to NYPA, or ask questions in full sentences. Hybrid search runs keyword matching for precision and semantic vector matching for meaning, and expands each query with synonyms, acronyms and

related concepts from NYPA's taxonomy and ontology. Administrators can tune and explain the resulting ranking, and test changes safely before they reach Production. This keeps search quality steadily improving and measurable rather than a black box.

Detailed requirements

- **SRCH-02.1** The Platform must execute every query against both a keyword index and a semantic vector index and combine the results using a documented fusion method (for example, reciprocal rank fusion or a weighted linear combination).
- **SRCH-02.2** The Platform must expand queries using the managed taxonomy and ontology: synonyms, preferred and alternative labels, acronyms, and, where configured, narrower and related terms.
- **SRCH-02.3** Items tagged with a taxonomy term that matches the query concept must receive a configurable relevance boost.
- **SRCH-02.4** Administrators must be able to tune relevance without code: field weights, freshness decay, source weights, content-type weights, authoritative-source boosts and the keyword-to-semantic balance.
- **SRCH-02.5** Administrators must be able to define promoted results ("best bets") for specific queries or query patterns, with start and end dates and audience targeting.
- **SRCH-02.6** The Platform must provide a relevance explanation for administrators showing why a result ranked where it did (keyword score, semantic score, taxonomy boost, graph boost, freshness).
- **SRCH-02.7** Relevance configuration changes must be testable in the Dev/Test/QA environment against a saved query set before promotion to Production.
- **SRCH-02.8** Search must support natural language queries of up to 500 characters.

Acceptance criteria

1. A query using an acronym (for example, "LOTO") returns documents that use only the expanded form ("lockout tagout").
2. A query using a synonym defined in the taxonomy returns content tagged with the preferred term.
3. On a NYPA-approved evaluation set of at least 100 queries with judged relevant documents, hybrid search achieves a higher normalized discounted cumulative gain at 10 (nDCG@10) than keyword-only search.
4. An administrator changes a field weight in Dev/Test/QA, runs the saved query set, compares before and after rankings, and promotes the change.

SRCH-03 Faceted navigation based on taxonomy and ontology

Priority	Mandatory
RFP reference	2.1 Unified Access, Search and Navigation, item 4

Description. Faceted navigation lets users narrow a large result set by clicking on meaningful categories such as facility, asset class, topic, document type and lifecycle status. These categories come from NYPA's controlled taxonomy and ontology rather than from inconsistent free-text fields. Hierarchical facets follow the structure of the taxonomy, so selecting a region or facility includes everything beneath it. A browse mode lets users explore the knowledge base by topic even when they do not know what to search for.

Detailed requirements

- **SRCH-03.1** Search results must be filterable by facets derived from the managed taxonomy and ontology, such as business unit, department, site or facility, asset or equipment class, topic, document type, knowledge asset type, lifecycle status, owner and date range.
- **SRCH-03.2** Hierarchical facets must display the term hierarchy and allow selection at any level, with selection of a parent term including all descendants.
- **SRCH-03.3** Facet values must show counts that reflect only content the user is permitted to see.
- **SRCH-03.4** Users must be able to combine multiple facet values (OR within a facet, AND across facets) and clear any selection.
- **SRCH-03.5** Administrators must be able to choose which facets appear, their order, their display labels and their default expansion, per landing page variant or search vertical.
- **SRCH-03.6** The Platform must provide a browse experience, independent of a query, that lets users navigate content by taxonomy hierarchy.
- **SRCH-03.7** Facet values must display preferred labels from the taxonomy, not internal identifiers.

Acceptance criteria

1. Selecting a parent facility term returns content tagged with any of its child terms.
2. Facet counts for a restricted user exclude items that user cannot access.
3. An administrator adds a new taxonomy-driven facet and it appears in search without code changes.

SRCH-04 Search verticals and scopes

Priority	Mandatory
RFP reference	2.1 Unified Access, Search and Navigation; Personalization and Role-Aware Experience

Description. Different user groups search for different kinds of knowledge. Field engineers look for procedures and drawings, analysts for policies and reports, and Canals staff for Canals-specific operating material. Search verticals are pre-configured search experiences with their own scope, filters and result layouts, targeted to the audiences that need them. A People vertical helps users find colleagues with relevant expertise, which supports NYPA's aim of making institutional knowledge easier to reach.

Detailed requirements

- **SRCH-04.1** Administrators must be able to define search verticals (for example, All, Procedures, Policies, Engineering Drawings, Lessons Learned, People, Canals) each with its own source scope, filters, facets and result layout.
- **SRCH-04.2** Verticals must be assignable to audiences so that role-relevant verticals appear by default.
- **SRCH-04.3** The Platform should provide a People vertical that returns colleagues by expertise, derived from authorship and taxonomy tags, subject to directory visibility rules.

Acceptance criteria

1. An administrator creates a Procedures vertical limited to procedure content types, and it appears for the assigned audience.

2.4 Repository connectors and permission synchronization

CONN-01 SharePoint Online connector

Priority	Mandatory
RFP reference	2.1 Unified Access, Search and Navigation, item 2; 2.4 EKP Data, Integration and Analytics; 2.1 Integration and Authoring

Description. SharePoint Online is NYPA's mandatory knowledge repository and the primary source of documents, pages and lists in the Platform. The SharePoint connector discovers and indexes content, metadata, managed metadata and permissions at every level, and keeps the index current through incremental change detection rather than repeated full crawls. It must operate within Microsoft's API limits and use least-privilege credentials. It must also honor NYPA's sensitivity labels, so that content NYPA designates as not indexable never enters the Platform.

Detailed requirements

- **CONN-01.1** The connector must index SharePoint Online document libraries, lists, list items, modern and classic site pages, news posts and wiki pages, and designated OneDrive locations, through Microsoft Graph and SharePoint REST APIs.
- **CONN-01.2** The connector must retrieve file content, standard and custom column metadata, managed metadata (term store) values, version history references, author and modifier, and item-level, folder-level, library-level and site-level permissions, including SharePoint groups, Entra ID security groups, Microsoft 365 groups and sharing links.
- **CONN-01.3** Administrators must be able to include or exclude site collections, sites, libraries, folders and content types.
- **CONN-01.4** The connector must perform an initial full crawl and then incremental synchronization using change tokens or delta queries, so that content, metadata and permission changes are reflected within the latency defined in NFR-PERF-04.
- **CONN-01.5** Deleted items in SharePoint must be removed from the index, and from answers and recommendations, within the latency defined in NFR-PERF-04.
- **CONN-01.6** The connector must respect Microsoft Graph throttling, using back-off and retry, without data loss.
- **CONN-01.7** The connector must authenticate using an Entra ID application registration with certificate-based credentials and least-privilege permissions (for example, Sites.Selected where practicable).
- **CONN-01.8** The connector must honor SharePoint sensitivity labels and must exclude content carrying labels NYPA designates as not indexable.

Acceptance criteria

1. A full crawl of a NYPA test site collection indexes all in-scope items with content, metadata and permissions.
2. Adding, editing, moving, deleting and changing permissions on test items are each reflected in search within the defined latency.
3. A label-excluded document never appears in the index.

CONN-02 Permission synchronization and security trimming

Priority	Mandatory
RFP reference	2.1 item 2 (“respects original permissions”); 2.1 Conversational Query and Answering item 2; 2.4 “permission-aware indexing”

Description. Permission synchronization is what lets the Platform respect original repository permissions, as the RFP requires. For each item, the Platform captures who may access it in the source system, keeps that information current, and applies it inside the search engine before any result is ranked or returned. The same enforcement covers AI answers, recommendations, facets and every other output. Where permissions cannot be confirmed, the Platform hides the item rather than risk exposure. Administrators can diagnose why any item is or is not visible to any user.

Detailed requirements

- **CONN-02.1** For every indexed item, the Platform must store the access control list in a normalized form (users and groups with allow and, where the source supports it, deny entries).
- **CONN-02.2** The Platform must resolve group membership, including nested groups, from Entra ID and from source-system local groups, and keep memberships current.
- **CONN-02.3** At query time, the Platform must compute the user’s effective security identifiers and apply them as a filter inside the search engine before results are ranked and returned, so that unauthorized items never leave the index.
- **CONN-02.4** The same filtering must be applied to answer retrieval, recommendations, trending lists, notifications, facets, counts, type-ahead suggestions, previews, thumbnails and knowledge graph traversals.
- **CONN-02.5** The Platform must support a configurable early-binding plus late-binding model in which, for high-sensitivity sources designated by NYPA, a real-time permission check against the source system is performed before an item is displayed.
- **CONN-02.6** If permission information for an item cannot be determined, the item must be treated as inaccessible (fail closed).
- **CONN-02.7** Administrators must be able to view, for any item, its synchronized permissions and the time they were last synchronized, and for any user, an explanation of why a given item is or is not visible.

Acceptance criteria

1. A permission test matrix covering direct grants, group grants, nested groups, sharing links, broken inheritance and removed access passes for search, answers, recommendations and facets with zero unauthorized disclosures.
2. Removing a user from a group results in loss of access to affected items within the latency defined in NFR-PERF-04.
3. Simulating a missing ACL for an item results in that item being hidden from all users except Platform Administrators using the diagnostic view.

CONN-03 OpenText Content Server connector

Priority	Target capability
RFP reference	2.1 item 2 (Content Server, target capability); 2.4 EKP

Description. OpenText Content Server holds a substantial body of NYPA records and controlled documents. Indexing it alongside SharePoint gives employees a single search across both. The connector retrieves documents, categories, attributes, versions and Content Server permissions, maps Content Server identities to NYPA directory identities, and keeps the index synchronized. The RFP identifies this as a target capability. It is built on the same connector framework as SharePoint so that behavior and administration are consistent.

Detailed requirements

- **CONN-03.1** The connector must index Content Server documents, folders, compound documents, categories and attributes, and version information through supported Content Server REST APIs.
- **CONN-03.2** The connector must retrieve Content Server permissions (owner, owner group, public access and assigned access) and map Content Server users and groups to NYPA directory identities.
- **CONN-03.3** The connector must support incremental synchronization and deletion handling equivalent to CONN-01.4 and CONN-01.5.
- **CONN-03.4** Content Server category attributes must be mappable to Platform metadata fields and taxonomy terms (see DES-04).

Acceptance criteria

1. In-scope Content Server test content is searchable alongside SharePoint content, with permissions enforced per the permission test matrix.

CONN-04 ServiceNow knowledge connector

Priority	Target capability
RFP reference	2.1 item 2 (ServiceNow, target capability); 2.4 EKP

Description. ServiceNow knowledge bases contain IT and operational support articles that employees often need alongside policy and engineering content. The connector indexes published articles with their attachments and categorization, and enforces ServiceNow user criteria so that article-level restrictions are respected. Only published articles appear by default, so draft and retired material does not surface in search or answers. The RFP identifies this as a target capability.

Detailed requirements

- **CONN-04.1** The connector must index ServiceNow knowledge base articles, including article body, attachments, knowledge base, category, workflow state, validity date and author.
- **CONN-04.2** The connector must enforce ServiceNow user criteria (“can read” and “cannot read”) and knowledge base access, either by synchronizing the criteria or by late-binding checks.
- **CONN-04.3** Only articles in the published workflow state must be indexed unless NYPA configures otherwise.
- **CONN-04.4** The connector should optionally index additional ServiceNow tables NYPA designates (for example, catalog items) as a configurable extension.

Acceptance criteria

1. Published ServiceNow test articles appear in unified search and answers; retired and draft articles do not; user criteria are enforced.

CONN-05 Intranet and additional repository connectors

Priority	Target capability
RFP reference	2.4 EKP (“M365, Content Server, ServiceNow, and intranet repositories”)

Description. NYPA’s intranet and other repositories also hold knowledge that should be findable from one place. A web connector indexes intranet pages, including Sitecore-delivered content. A documented connector development kit and ingestion API allow additional repositories to be connected later without changing the core product. Optional Microsoft Teams and Stream connectors extend coverage to collaboration files and recorded video in Microsoft 365.

Detailed requirements

- **CONN-05.1** The Platform must provide a web connector able to index NYPA intranet sites (including Sitecore-delivered pages) with configurable seed URLs, include and exclude patterns, crawl depth and authentication.
- **CONN-05.2** The Platform must provide a documented connector software development kit or ingestion API so that additional repositories can be connected without changes to the core product.
- **CONN-05.3** The Platform should provide connectors for Microsoft Teams channel files and Microsoft Stream video in Microsoft 365.

CONN-06 Connector administration and monitoring

Priority	Mandatory
RFP reference	2.4 EKP; Section 6 Architecture Guideline 3

Description. Connectors are critical infrastructure: if one stops working, search results become stale or incomplete without users noticing. This requirement gives NYPA administrators full control and visibility over every connector, including configuration, scheduling, run status, item-level errors and throughput. Failures generate proactive alerts. Connectors are standard, vendor-certified and configurable rather than custom code, in line with NYPA's architecture guideline on application connectors.

Detailed requirements

- **CONN-06.1** Administrators must be able to create, configure, schedule, pause, resume and run each connector from the administrative console.
- **CONN-06.2** For each connector, the console must show status, last successful run, items discovered, indexed, updated, deleted and failed, error details per item, and throughput.
- **CONN-06.3** The Platform must alert administrators by email and Teams when a connector fails, falls behind its latency target, or encounters an authentication error.
- **CONN-06.4** Failed items must be retried automatically and be re-runnable on demand.
- **CONN-06.5** Connectors must be vendor-certified and configurable rather than custom-coded per deployment.

Acceptance criteria

1. Revoking a connector credential produces an alert within 15 minutes and a visible error state in the console.

2.5 Personalization and role-aware experience

PERS-01 Role-based discovery

Priority	Mandatory
RFP reference	2.1 Personalization and Role-Aware Experience, item 1

Description. Role-based discovery tailors what the Platform puts in front of each user, using who they are (department, job title, business unit, location) and what they have said they care about. Administrators target experiences such as landing pages, verticals and featured content to audiences defined from directory attributes and groups. Personalization only re-orders permitted content; it never hides relevant results. Users can see and adjust what drives their personalization, or turn it off.

Detailed requirements

- **PERS-01.1** The Platform must derive a user profile from the directory (department, job title, business unit, location, manager) and from explicit user preferences (topics of interest, preferred sources).

- **PERS-01.2** Administrators must be able to define audience rules based on profile attributes and directory groups, and target landing page variants, verticals, promoted results, featured content and topic hubs to those audiences.
- **PERS-01.3** Search ranking must include a configurable, bounded boost for content relevant to the user's department, facility and followed topics, without hiding other relevant content.
- **PERS-01.4** Users must be able to view and edit the interests the Platform uses to personalize their experience, and to turn personalization off.

Acceptance criteria

1. Two users from different departments issuing the same broad query see the same set of permitted results, with department-relevant items ranked higher for each.
2. A user turns personalization off and receives unpersonalized ranking.

PERS-02 Personalized homepage

Priority	Mandatory
RFP reference	2.1 Personalization item 1

Description. The personalized homepage is each employee's own workspace in the Platform. It gathers recommended content, updates to items they watch, new content in topics they follow, alerts from saved searches, recently viewed items and assigned tasks. It gives employees a reason to return to the Platform daily and turns it from a search tool into a working environment. New users see sensible role-based defaults from their first visit.

Detailed requirements

- **PERS-02.1** Each user must have a personalized homepage that shows, at minimum: recommended content, updates to watched items, new content in followed topics, saved searches with new result counts, recently viewed items and assigned tasks.
- **PERS-02.2** Users must be able to show, hide and reorder homepage components within the bounds the administrator allows.
- **PERS-02.3** The homepage must display meaningful content on a user's first visit, using role and department defaults until personal signals accumulate.

PERS-03 Topic hubs

Priority	Mandatory
RFP reference	2.1 Personalization item 1

Description. Topic hubs are pages dedicated to a subject that matters to NYPA, such as dam safety, transmission maintenance or a strategic program. Each hub is driven by the taxonomy, so it fills automatically with content tagged with the topic and its narrower terms. Topic owners can curate it with featured items and context. A hub also brings together related topics, recognized experts and trending items, so employees can get up to speed on a subject quickly.

Detailed requirements

- **PERS-03.1** The Platform must provide a topic hub for any taxonomy term designated as a hub, automatically aggregating content tagged with the term and its narrower terms.
- **PERS-03.2** Each topic hub must show: a description of the topic, curated featured content, the most relevant and most recent content, related topics from the ontology and knowledge graph, recognized experts, trending items, and a scoped search and question box.
- **PERS-03.3** Topic owners must be able to curate the hub (featured items, description, pinned links) without administrator involvement.
- **PERS-03.4** Users must be able to follow a topic hub and receive notifications of new or changed content.

Acceptance criteria

1. Designating a taxonomy term as a hub creates a populated hub page without manual content assignment.
2. A topic owner curates featured items, and a follower is notified of a new tagged document.

PERS-04 Saved searches

Priority	Mandatory
RFP reference	2.1 Personalization item 1

Description. Saved searches let users keep a search they run often, including its filters and scope, and rerun it with one click. Subscribing to a saved search turns it into a standing alert, so users are told when new matching content appears rather than having to check repeatedly. Alerts are permission-filtered when they are generated, so they never disclose content the recipient cannot access.

Detailed requirements

- **PERS-04.1** Users must be able to save any search, including the query, selected facets, scope and sort order, and give it a name.
- **PERS-04.2** Users must be able to rerun, rename, edit and delete saved searches.
- **PERS-04.3** Users must be able to subscribe to a saved search and receive alerts of new matching results at a chosen frequency (immediate, daily or weekly) by email, Teams or in-app notification.
- **PERS-04.4** Alerts must only include items the user is permitted to see at the time the alert is generated.

PERS-05 Watchlists

Priority	Mandatory
RFP reference	2.1 Personalization item 1

Description. Watchlists let users track specific items, topics or people and be notified when they change: a new version, a status change such as being superseded or retired, or new related content. This is particularly important for procedures and policies, where using an outdated version carries operational and compliance risk. Users organize watchlists and control how and when they are notified.

Detailed requirements

- **PERS-05.1** Users must be able to add any document, page, article, topic or person to a watchlist.
- **PERS-05.2** Users must be notified when a watched item is updated, has a new version, changes lifecycle state (for example, superseded or retired) or receives a new related item.
- **PERS-05.3** Users must be able to manage watchlists, including grouping items into named lists and setting notification preferences per list.

PERS-06 Recommendations

Priority	Mandatory
RFP reference	2.1 Personalization item 2

Description. Recommendations surface relevant knowledge the user did not search for. They are based on content similarity, shared taxonomy topics, knowledge graph relationships and the activity of colleagues in similar roles. Every recommendation explains why it was made, which builds trust and helps users judge relevance. User feedback, such as dismissing an unhelpful recommendation, improves future suggestions. Outdated, superseded and retired content is never recommended.

Detailed requirements

- **PERS-06.1** The Platform must recommend content to each user based on a combination of content similarity to items they have used, shared taxonomy tags with their interests, knowledge graph proximity, and the behavior of colleagues with similar roles.
- **PERS-06.2** On every document preview the Platform must show related content (“more like this”) and, where applicable, newer versions and items that supersede it.
- **PERS-06.3** Each recommendation must display a short reason (for example, “Because you follow Dam Safety” or “Popular in Transmission Engineering”).
- **PERS-06.4** Users must be able to dismiss a recommendation and indicate it is not relevant; this feedback must influence future recommendations.
- **PERS-06.5** Recommendations must exclude superseded, retired and archived content.

Acceptance criteria

1. A new user in a given department receives department-relevant recommendations on first sign-in.

2. Recommendation reasons are displayed and dismissals are reflected in subsequent recommendations.

PERS-07 Trending content in users' domains

Priority	Mandatory
RFP reference	2.1 Personalization item 2

Description. Trending content shows what colleagues in the same domain are reading and searching for now, for example during an outage, a regulatory change or a seasonal operation. Trending is calculated at organization, business unit, department and topic level, favors recent activity, and is adjusted so that items that are always popular do not crowd out emerging ones. Each user sees trends for their own domain, filtered by permission.

Detailed requirements

- **PERS-07.1** The Platform must compute trending content and trending queries over configurable windows (for example, 7 and 30 days) at organization, business unit, department and topic levels.
- **PERS-07.2** Trending calculations must weight recent activity more heavily than older activity and normalize for overall item popularity.
- **PERS-07.3** Trending lists shown to a user must be scoped to the user's domain (department, facility or followed topics) and filtered by permission.

2.6 Conversational query and answering

CQA-01 Natural language answers alongside search results

Priority	Mandatory
RFP reference	2.1 Conversational Query and Answering, item 1

Description. Conversational answering lets employees ask a question in plain language and receive a direct answer, rather than opening and reading several documents. The answer appears alongside the traditional search results, so users can confirm it against the sources or continue browsing. A dedicated assistant view supports longer, multi-turn conversations. Answers follow the user's current scope, so a question asked within a topic hub or vertical is answered from that context.

Detailed requirements

- **CQA-01.1** When a user enters a question in the search box, the Platform must return a direct natural language answer above or beside the traditional ranked search results on the same page.
- **CQA-01.2** The Platform must detect whether input is a question or a keyword query and should show an answer only when an answer is likely to be useful; users must be able to request an answer explicitly.
- **CQA-01.3** The Platform must also provide a dedicated conversational assistant view for multi-turn conversations.
- **CQA-01.4** Answers must stream progressively so that users see the first words quickly (see NFR-PERF-02).

- **CQA-01.5** Answers must be formatted for readability, including numbered steps, bullet lists and tables where the source content warrants.
- **CQA-01.6** Users must be able to copy an answer, share it (with citations), and provide feedback (helpful or not helpful, with an optional reason).
- **CQA-01.7** Answer scope must follow the active search scope and facets, so that a user can ask a question within a vertical, a topic hub or a single document.

Acceptance criteria

1. A natural language question returns an answer and ranked results on the same page.
2. A question asked within a topic hub is answered only from content in that hub.

CQA-02 Retrieval-augmented, cited and permission-aware answers

Priority	Mandatory
RFP reference	2.1 Conversational Query and Answering, item 2

Description. Answers must be trustworthy enough for operational use. Every answer is generated only from NYPA content retrieved at the time of the question (retrieval-augmented generation), not from the language model's general knowledge. Retrieval applies the user's permissions before any content reaches the model, so an answer can never draw on restricted material. Each statement links to the passage that supports it. Current, published and authoritative sources are preferred. When the knowledge base does not contain an answer, the Platform says so instead of guessing.

Detailed requirements

- **CQA-02.1** Every answer must be generated from content retrieved from the Platform's index at the time of the question; the language model must not answer from its general training knowledge alone.
- **CQA-02.2** Retrieval must apply the same permission filtering as search (CONN-02.3) before any content is passed to the language model.
- **CQA-02.3** Every factual statement in an answer must be supported by at least one inline citation linking to the specific source document and, where possible, the page, section or passage.
- **CQA-02.4** Users must be able to open each citation and see the cited passage highlighted.
- **CQA-02.5** Retrieval must prefer authoritative sources: published over draft, current over superseded, and sources designated authoritative by administrators.
- **CQA-02.6** When retrieved content does not support an answer, the Platform must say that it could not find an answer in the permitted knowledge base and offer the search results, rather than generate an unsupported answer.
- **CQA-02.7** When sources conflict, the answer must state the conflict and cite each source.
- **CQA-02.8** Administrators must be able to exclude specific sources, content types or sensitivity labels from answer generation while keeping them searchable.

Acceptance criteria

1. On a NYPA-approved evaluation set of at least 100 questions, at least 95 percent of answer statements are supported by their cited passages (citation precision), measured by reviewer judgment.
2. For questions whose answers exist only in content the test user cannot access, the Platform returns no answer content derived from that material, in 100 percent of test cases.
3. For questions with no answer in the corpus, the Platform declines to answer in at least 90 percent of cases.

CQA-03 Follow-up questions and conversational refinement

Priority	Mandatory
RFP reference	2.1 Conversational Query and Answering, item 3

Description. Real questions rarely end with one exchange. Users narrow, redirect and probe, for example “What about at Blenheim-Gilboa?” or “Show me the latest version”. The Platform keeps the context of the conversation so that follow-up questions are understood in light of what came before. Where a refinement matches a filter such as date, version, source or department, the Platform applies that filter to retrieval rather than relying on the language model to interpret it. Conversations can be resumed later and remain private to the user.

Detailed requirements

- **CQA-03.1** The Platform must maintain conversation context so that follow-up questions can refer to earlier questions and answers (for example, “What about for Blenheim-Gilboa?”).
- **CQA-03.2** The Platform must support refinement instructions, including “Show me the latest version”, “Only from Canals”, “Summarize that in three bullets”, “Who owns this?” and “What changed from the previous version?”
- **CQA-03.3** Refinements that correspond to filters (date, version, source, department, document type) must be applied as structured filters on retrieval, not only as text instructions to the language model.
- **CQA-03.4** The Platform must offer suggested follow-up questions based on the answer and the retrieved content.
- **CQA-03.5** Users must be able to start a new conversation, view their conversation history, rename and delete conversations, and resume a past conversation.
- **CQA-03.6** Conversation history must be private to the user, retained according to NYPA’s configured retention period, and deletable by the user.
- **CQA-03.7** Users must be able to convert an answer into a search with the inferred filters applied.

Acceptance criteria

1. “Show me the latest version” after an answer about a procedure returns an answer citing only the most recent published version.
2. A resumed conversation retains context from earlier turns.

CQA-04 Confidence scores and explainability

Priority	Mandatory
RFP reference	2.1 Conversational Query and Answering, item 4

Description. Users need to know how far to rely on an answer. Each answer carries a confidence indicator calculated from measurable signals: how relevant the retrieved sources were, whether they agree, how much of the answer is covered by citations, and how authoritative and current the sources are. An explanation panel shows exactly which sources, taxonomy expansions, graph relationships and filters were used. Low-confidence answers are clearly flagged. Every answer is recorded with its inputs and model version for quality review and audit.

Detailed requirements

- **CQA-04.1** Each answer must display a confidence indicator (for example, High, Medium, Low, with an underlying numeric score) computed from measurable signals such as retrieval relevance, agreement among sources, citation coverage of the answer statements, and source authority and freshness.
- **CQA-04.2** Users must be able to open an explanation panel that shows: the sources retrieved and used, their relevance, the taxonomy terms and graph relationships used to expand the question, filters applied, and why low confidence was assigned where applicable.
- **CQA-04.3** Answers below an administrator-configured confidence threshold must carry a visible caution and a prompt to verify against the source.
- **CQA-04.4** The confidence model must be calibrated against the evaluation set so that higher displayed confidence corresponds to higher measured accuracy.
- **CQA-04.5** The Platform must record, for every answer, the question, retrieved items, model and prompt version, confidence score and user feedback, for quality analysis and audit (see NFR-AI-03).

Acceptance criteria

1. On the evaluation set, answers labeled High confidence are correct more often than those labeled Medium, which are correct more often than those labeled Low.
2. The explanation panel lists every source passed to the model and the expansions applied.

2.7 Knowledge content coverage

COV-01 Multiple knowledge asset formats

Priority	Mandatory
RFP reference	2.1 Knowledge Content Coverage, item 1

Description. NYPA knowledge exists in many forms: Office documents, PDFs (many of them scanned), presentations, spreadsheets, emails, images, engineering drawings and recorded video and audio. The Platform extracts searchable text from all of these. It uses optical character recognition for scanned material and transcription for recordings, and indexes long items by page, slide, section or time offset. As a result, search results and AI answers can point to the exact place in a document or recording where the relevant knowledge appears.

Detailed requirements

- **COV-01.1** The Platform must extract and index text from at least the following formats: Microsoft Word (DOC, DOCX), PowerPoint (PPT, PPTX), Excel (XLS, XLSX, CSV), PDF (including scanned PDFs), RTF, TXT, Markdown, HTML, XML, MSG and EML email files, Visio (VSDX) and OneNote pages.
- **COV-01.2** The Platform must apply optical character recognition to scanned PDFs and to images (PNG, JPEG, TIFF, BMP, GIF) so that text within them is searchable.
- **COV-01.3** The Platform must index video and audio files (at minimum MP4, MOV, WEBM, M4A, MP3, WAV) by generating a time-coded transcript, so that spoken content is searchable and citable.
- **COV-01.4** Search results and citations for video and audio must link to the relevant time offset.
- **COV-01.5** For images, the Platform should generate a descriptive caption to support search and accessibility.
- **COV-01.6** For presentations and long documents, the Platform must index at slide or section granularity so that answers and results can point to the specific slide or section.
- **COV-01.7** Engineering formats NYPA designates (for example, DWG drawings) must at minimum be indexed by file name and metadata, with title block text extraction where feasible.
- **COV-01.8** Password-protected or encrypted files that cannot be read must be indexed by metadata only and reported on the content health dashboard.

Acceptance criteria

1. A reference set containing every listed format is ingested, and a known phrase from each file is found by search.
2. A phrase spoken in a test video is found by search, and the result opens the video at the correct time.

COV-02 Site pages, wikis and knowledge articles in connected systems

Priority	Mandatory
RFP reference	2.1 Knowledge Content Coverage, item 2

Description. Much of NYPA's knowledge is published as SharePoint site pages, news posts and wiki pages, ServiceNow knowledge articles and intranet pages rather than as files. The Platform indexes these page-based formats with their content and metadata, presents them with page-appropriate result layouts, and records the links between pages and documents in the knowledge graph. This lets page content and document content be found and connected together.

Detailed requirements

- **COV-02.1** The Platform must index SharePoint modern site pages, news posts and classic wiki pages, including web part text content and page metadata.
- **COV-02.2** The Platform must index ServiceNow knowledge articles and intranet pages when the respective connectors are enabled.
- **COV-02.3** Page-type content must be presented with a page-appropriate result template showing site, page title, summary and last update.
- **COV-02.4** Links between pages and documents must be captured as relationships in the knowledge graph.

COV-03 Indexing, previews and thumbnails

Priority	Mandatory
RFP reference	2.1 Knowledge Content Coverage, item 3

Description. Users decide faster whether a result is useful if they can see it without downloading it. The Platform provides in-browser previews for common formats, with search terms or cited passages highlighted. It generates thumbnails for documents, images and videos, and provides a video and audio player with a synchronized, searchable transcript. Previews show the item's metadata, ownership, lifecycle status and related items. They are permission-checked and never cached in a way that could expose content.

Detailed requirements

- **COV-03.1** The Platform must provide an in-browser preview for all common document formats (Office, PDF, images, text) without requiring a download or a desktop application.
- **COV-03.2** The preview must highlight the search terms or the cited passage and allow navigation between matches.
- **COV-03.3** The Platform must generate thumbnails for documents (first page or first slide), images and videos (representative frame), and display them in search results.
- **COV-03.4** Video and audio previews must provide a player with the transcript displayed alongside, searchable within the transcript, with click-to-seek.
- **COV-03.5** Previews and thumbnails must be security-trimmed and must not be cached in a manner that bypasses permission checks (see NFR-SEC-08).
- **COV-03.6** The preview panel must show the item's metadata, taxonomy tags, owner, lifecycle status, version, related items and a link to open in the source system.

COV-04 Configurable templates for common knowledge asset types

Priority	Mandatory
RFP reference	2.1 Knowledge Content Coverage, item 4

Description. Consistent structure makes knowledge easier to create, review and reuse. Administrators define knowledge asset types, such as Standard Operating Procedure, How-To Guide, FAQ, Key Takeaways, Topic Guide and Policy, each with its own section structure, mandatory metadata, review interval, workflow and display template. Contributors create new assets from these templates and the files are stored in Microsoft 365. The Platform includes a starter library that NYPA can adapt, and template changes are versioned.

Detailed requirements

- **COV-04.1** Administrators must be able to define knowledge asset types (for example, Standard Operating Procedure, How-To Guide, FAQ, Key Takeaways, Topic Guide, Policy, Lessons Learned, Technical Reference, Training Material) without code.
- **COV-04.2** Each asset type must define: a content structure (required and optional sections with guidance text), mandatory and optional metadata fields, allowed taxonomy term sets per field, default review interval,

applicable publishing and review workflow, default audience, and display template for search results and previews.

- **COV-04.3** Contributors must be able to create a new asset from a template within the Platform; the resulting document must be stored in the designated Microsoft 365 location (see INT-01).
- **COV-04.4** The Platform must ship with a starter library of templates for the asset types listed in COV-04.1 that NYPA can adapt.
- **COV-04.5** Template changes must be versioned; existing assets must record which template version they were created from.
- **COV-04.6** The Platform should use the template structure to guide AI-generated summaries and completeness checks for that asset type.

Acceptance criteria

1. An administrator defines a new “Outage Debrief” asset type with sections, mandatory fields, a review interval and a workflow, and a contributor creates an asset from it.

2.8 Content enrichment and semantic layer integration

SEM-01 Managed metadata with hierarchical terms and synonyms

Priority	Mandatory
RFP reference	2.1 Content Enrichment and Semantic Layer Integration, item 1

Description. Managed metadata ensures content is described with a shared, controlled vocabulary rather than inconsistent free text. Term sets support deep hierarchies, synonyms, acronyms, definitions and ontology relationships beyond simple parent and child. Terms keep stable identifiers through relabeling, so renaming a term never requires retagging content. When terms are merged or deprecated, existing tags can be remapped in bulk with a full audit record.

Detailed requirements

- **SEM-01.1** The Platform must support managed metadata fields whose values are drawn from controlled term sets.
- **SEM-01.2** Term sets must support unlimited hierarchical depth (parent and child terms), preferred labels, alternative labels (synonyms), acronyms, hidden labels for common misspellings, definitions, and status (active, deprecated).
- **SEM-01.3** Terms must have stable identifiers that persist through label changes, so that relabeling a term does not require retagging content.
- **SEM-01.4** The Platform must support ontology relationships between terms beyond hierarchy, including “related to”, “part of”, “located at”, and NYPA-defined relationship types.
- **SEM-01.5** Metadata fields must support single-value and multi-value selection, and validation against the term set.

- **SEM-01.6** Users tagging content must be able to find terms by any label (preferred, alternative or acronym) through type-ahead.
- **SEM-01.7** When a term is deprecated or merged, the Platform must support remapping existing tags to the replacement term in bulk, with an audit record.
- **SEM-01.8** Taxonomy Managers must be able to maintain term sets in the Platform, subject to the synchronization rules in SEM-02 when the enterprise taxonomy tool is the master.

Acceptance criteria

1. A three-level facility hierarchy with synonyms is configured; tagging by synonym stores the preferred term; search by synonym retrieves the content.
2. Relabeling a term updates its display everywhere without retagging.
3. Merging two terms remaps all tagged content and records the change.

SEM-02 Integration with NYPA's taxonomy and ontology tool

Priority	Mandatory
RFP reference	2.1 Content Enrichment and Semantic Layer Integration, item 2; 2.2 Taxonomy/ Ontology Model Management

Description. NYPA already maintains controlled vocabularies in an enterprise taxonomy or ontology tool. The Platform must use those vocabularies rather than create a competing set. The Platform imports and synchronizes taxonomies using open standards such as SKOS and OWL, and treats the enterprise tool as master for the term sets NYPA designates. It keeps the SharePoint Online term store aligned so that SharePoint columns and the Platform use the same terms. Usage statistics and term suggestions flow back to the enterprise tool for governance.

Detailed requirements

- **SEM-02.1** The Platform must import taxonomies and ontologies from NYPA's current taxonomy tool using open standards, at minimum SKOS (RDF/XML, Turtle or JSON-LD) and CSV, and should support OWL for ontology classes and properties.
- **SEM-02.2** The Platform must support scheduled and on-demand synchronization with the taxonomy tool through its API, treating the taxonomy tool as the master for designated term sets.
- **SEM-02.3** Synchronization must handle additions, label changes, moves within the hierarchy, deprecations and merges, and produce a change report.
- **SEM-02.4** The Platform must synchronize designated term sets with the SharePoint Online term store so that SharePoint managed metadata columns and Platform metadata use the same terms and identifiers.
- **SEM-02.5** The Platform must export its term sets and term usage statistics in SKOS and CSV so that the enterprise taxonomy tool can consume them.
- **SEM-02.6** Term suggestions generated by the Platform (see SEM-04) must be exportable to the taxonomy tool for governance rather than added directly to master term sets.

- **SEM-02.7** Synchronization failures must be reported to Taxonomy Managers with details and must not leave term sets in a partially updated state.

Acceptance criteria

1. A taxonomy exported from NYPA's tool is imported and used for tagging and facets.
2. A label change and a term move made in the taxonomy tool appear in the Platform after the next synchronization, without retagging.
3. A designated term set is consistent between the Platform and the SharePoint term store after synchronization.

SEM-03 Knowledge graph enrichment

Priority	Mandatory
RFP reference	2.1 Content Enrichment and Semantic Layer Integration, item 3; 2.2 Knowledge Graph

Description. The knowledge graph connects documents to the concepts, people, facilities, assets, projects and organizational units they relate to. This lets the Platform understand relationships, not just words. The graph is used in three ways, as the RFP requires. It boosts search results that are connected to entities in the query. It lets users navigate from a document to related assets, experts and procedures. It adds connected context to the passages used for conversational answers. The graph is filtered by permission and can be used by downstream NYPA data platforms and AI applications.

Detailed requirements

- **SEM-03.1** The Platform must maintain a knowledge graph whose nodes include documents, taxonomy terms, ontology classes, people, departments, facilities and sites, assets and equipment, projects and source systems, and whose edges include "tagged with", "authored by", "owned by", "references", "supersedes", "located at", "part of", "related to" and NYPA-defined relationships.
- **SEM-03.2** The graph must be populated automatically from taxonomy tags, source metadata, document links, entity extraction from content, and directory data, and must update as content changes.
- **SEM-03.3** Graph-enriched search: ranking must include a configurable boost for results connected to entities recognized in the query (for example, a query mentioning a facility boosts documents linked to that facility and its assets).
- **SEM-03.4** Graph-enriched navigation: users must be able to explore related entities from any document, topic or person, including "documents about this asset", "experts on this topic", "other procedures at this facility" and "what this supersedes".
- **SEM-03.5** The Platform should provide an interactive visual graph explorer for a selected entity, limited to a configurable number of hops and filtered by permission.
- **SEM-03.6** Graph-enriched conversational querying: answer retrieval must use graph relationships to expand and disambiguate questions and to add connected passages to the retrieval set; the explanation panel must show which relationships were used.

- **SEM-03.7** Graph traversals must never expose the existence, title or relationships of content the user cannot access.
- **SEM-03.8** The knowledge graph must be consumable by downstream data platforms and applications through a documented API and through bulk export in standard formats (for example, RDF or labeled property graph CSV).
- **SEM-03.9** The Platform should be able to integrate with an external enterprise knowledge graph that NYPA designates, consuming its entities and relationships rather than duplicating them.

Acceptance criteria

1. On the evaluation set, graph-enriched search improves nDCG@10 over hybrid search without graph boosting.
2. From a procedure, a user navigates to the facility, to the asset, to other documents about that asset, and to the subject matter expert.
3. A downstream test client retrieves graph data through the API using OAuth credentials.

SEM-04 AI-assisted vocabulary improvement

Priority	Mandatory
RFP reference	2.1 Content Enrichment and Semantic Layer Integration; 2.2 Taxonomy/Ontology Model Management (AI-assisted model development)

Description. Vocabularies need ongoing care as language, assets and organizational priorities change. The Platform analyzes real usage, such as queries that return no results, recurring phrases not yet in the taxonomy, and corrections reviewers make to tags. From these it proposes candidate terms, synonyms and relationships to taxonomy managers, with supporting evidence. Every suggestion requires human approval before it changes a master vocabulary.

Detailed requirements

- **SEM-04.1** The Platform must suggest candidate terms, synonyms and relationships based on frequent queries that return no or poor results, recurring phrases in content not covered by existing terms, and tagging corrections made by reviewers.
- **SEM-04.2** Suggestions must be presented to Taxonomy Managers in a review queue with supporting evidence (example queries, example documents, frequency), and be accepted, edited, rejected or exported.
- **SEM-04.3** No suggestion may modify a master term set without human approval.

2.9 Content ingestion

ING-01 Easy upload interface

Priority	Mandatory
RFP reference	2.1 Content Ingestion

Description. Contributing knowledge must be as easy as dropping a file. Users drag and drop single files, multiple files or whole folders, including large video recordings, and the Platform scans them for malware and starts processing them immediately. Within moments the contributor sees one review screen with a suggested summary, suggested tags, a completeness check and any duplicate warnings. Most of the effort of describing content therefore moves from the contributor to the Platform. Administrators can also run bulk ingestion with metadata mapping for migrations.

Detailed requirements

- **ING-01.1** Users must be able to upload content by drag and drop and by a file picker, from the landing page, a topic hub, or the contribute page.
- **ING-01.2** Users must be able to upload multiple files and folders in a single action, with per-file progress, pause and cancel, and resumable uploads for large files.
- **ING-01.3** The Platform must support individual files up to at least 5 GB, so that video recordings can be uploaded.
- **ING-01.4** Users must select or be offered a destination (a SharePoint library mapped to a department or topic) and an asset type template; the Platform should suggest a destination based on content and the user's department.
- **ING-01.5** The Platform must scan uploads for malware before they are stored or processed, and reject infected files with a clear message.
- **ING-01.6** The Platform must enforce administrator-configured file type and size rules.
- **ING-01.7** Immediately after upload, the Platform must start processing (text extraction, summary, classification, duplicate check, completeness check) and present results to the contributor in a single review screen (see META-01 to META-03).
- **ING-01.8** The Platform must warn the contributor at upload time if the file is an exact or near duplicate of existing content, with links to the existing items (see GOV-06).
- **ING-01.9** Contributors must be able to save a submission as a draft and return to it later.
- **ING-01.10** Administrators must be able to perform bulk ingestion from a file share or SharePoint location with a metadata mapping file.

Acceptance criteria

1. A user drags ten files of mixed formats into the upload area, receives suggested metadata for each, and submits them for review.
2. An EICAR test file is rejected.
3. A 3 GB video uploads successfully and is transcribed.

2.10 Automated metadata generation

META-01 AI-generated summary with human-in-the-loop review

Priority	Mandatory
RFP reference	2.1 Automated Metadata Generation, item 1

Description. A good short description helps users judge relevance from search results and helps reviewers understand a submission quickly, but few contributors write one. At ingest, the Platform generates a brief description from the full content of each item, including transcripts for recordings. The description is clearly labeled as AI-generated, and a person can accept, edit or reject it before publication. The Platform records how each published description was produced and who approved it. Descriptions are regenerated when content changes substantially.

Detailed requirements

- **META-01.1** At ingest, the Platform must generate a brief description of each document (configurable length, default two to four sentences) from its full content, including transcripts for video and audio.
- **META-01.2** The generated description must be presented to the contributor or steward as a suggestion, clearly labeled as AI-generated, that can be accepted, edited or rejected before publication.
- **META-01.3** The Platform must record whether the published description was AI-generated, AI-generated and edited, or human-written, and who approved it.
- **META-01.4** For documents ingested by connectors (not uploaded), administrators must be able to configure whether generated descriptions are applied automatically, applied pending steward review, or held in a review queue.
- **META-01.5** When a document is substantially updated, the Platform must regenerate the description and route it for review according to configuration.
- **META-01.6** Generated descriptions must be written back to a designated metadata column in SharePoint when configured.

Acceptance criteria

1. On a sample of 100 documents, reviewers rate at least 85 percent of generated descriptions as accurate and usable with no or minor edits.
2. Edit and approval history for a description is visible in the item’s audit trail.

META-02 Auto-classification and auto-tagging with human-in-the-loop review

Priority	Mandatory
RFP reference	2.1 Automated Metadata Generation, item 2; 2.2 Auto-Classification and Tagging

Description. Manual tagging is slow and inconsistent, and it is often skipped. The Platform automatically classifies each item by asset type and assigns taxonomy terms to each managed metadata field, with a confidence score for every suggestion. High-confidence tags are applied automatically and lower-confidence tags are routed to a person for review. Thresholds are configurable per field. Reviewer corrections improve future classification. Classification can be triggered on upload or update, on a schedule, or manually, and runs at a scale of at least 500 documents in 30 minutes.

Detailed requirements

- **META-02.1** At ingest and on update, the Platform must automatically classify each item by asset type and assign taxonomy terms to each managed metadata field, using the content, the existing metadata and the term definitions and synonyms.
- **META-02.2** Each suggested tag must carry a confidence score. Administrators must be able to set, per field, a threshold above which tags are applied automatically and below which tags are routed for review.
- **META-02.3** Contributors and stewards must be able to accept, reject or replace each suggested tag in the review screen, and add tags manually.
- **META-02.4** Reviewer corrections must be recorded and used to improve future classification (for example, as examples for few-shot classification or for retraining), and must be visible in classification quality reporting.
- **META-02.5** The Platform must support classification triggered by upload or update, by scheduled batch over a selected scope, and by manual initiation for a selected item or set.
- **META-02.6** The Platform must classify at least 500 documents within 30 minutes in a batch job.
- **META-02.7** The Platform must provide monitoring of classification jobs, including items processed, auto-applied, routed for review, failed, and acceptance rate per field over time.
- **META-02.8** The Platform must extract named entities (people, facilities, assets, equipment identifiers, regulations, projects) and map them to taxonomy terms or graph entities where a match exists.
- **META-02.9** Confirmed tags must be written back to SharePoint managed metadata columns when configured.

Acceptance criteria

1. On a NYPA-labeled reference set of at least 200 documents, auto-applied tags achieve at least 85 percent precision per field at the configured threshold.
2. A batch of 500 documents completes classification within 30 minutes.
3. A reviewer’s correction is recorded and reflected in the field’s acceptance rate.

META-03 Minimum metadata completeness before publishing

Priority	Mandatory
RFP reference	2.1 Automated Metadata Generation, item 3

Description. Search, navigation and governance only work if content is properly described. Administrators set completeness rules per asset type, including mandatory fields, minimum tag counts, format rules and a weighted completeness score. The Platform blocks publication until the threshold is met. Contributors and reviewers see what is missing as they work. Authorized administrators can grant recorded exceptions where necessary, and completeness across the whole corpus is reported on the content health dashboard.

Detailed requirements

- **META-03.1** Administrators must be able to define completeness rules per asset type, consisting of mandatory fields, minimum counts for multi-value fields (for example, at least one topic tag), field format rules, and a weighted completeness score with a minimum threshold.
- **META-03.2** The Platform must calculate and display the completeness score and list missing or invalid fields during authoring and review.
- **META-03.3** The Platform must block transition to the published state when an item is below the threshold or missing a mandatory field.
- **META-03.4** Administrators with the appropriate permission may grant a recorded exception for a specific item, with a reason.
- **META-03.5** The Platform must report completeness across the corpus by source, department and asset type on the content health dashboard, including connector-ingested content that predates the rules.

Acceptance criteria

1. An item missing a mandatory field cannot be published, and the reason is shown to the submitter.
2. An exception is granted, recorded with a reason and visible in the audit trail.

2.11 Governance and content lifecycle management

GOV-01 Configurable publishing workflows

Priority	Mandatory
RFP reference	2.1 Governance and Content Lifecycle Management, item 1

Description. Publishing workflows ensure that knowledge is reviewed and approved by the right people before employees rely on it. Administrators configure the stages, approvers, approval rules, due dates and escalations for each asset type, department or location, without code. Approvers work from a task queue that shows the submission, the AI suggestions accepted or changed, completeness status and changes from the previous version. Authors cannot approve their own work. Every decision is recorded.

Detailed requirements

- **GOV-01.1** The Platform must provide a publishing workflow with, at minimum, the stages Draft, Submitted, In Review, Approved and Published, plus Rejected and Returned for Changes.
- **GOV-01.2** Administrators must be able to configure workflows without code: add, remove, rename and reorder stages; define parallel or sequential approvals; require one, any or all approvers at a stage; and set stage due dates and escalation.
- **GOV-01.3** Approvers must be assignable by role, by scope (department, topic, business unit), by named user or group, or by a metadata-driven rule (for example, the owner of the topic tagged).
- **GOV-01.4** Different workflows must be assignable per asset type, department or source location.
- **GOV-01.5** The author of an item must not be able to approve their own submission unless an administrator explicitly allows it for a workflow.
- **GOV-01.6** Each approver must see a task queue with item details, the AI suggestions accepted or edited, completeness status, duplicate warnings and a comparison with the previous version.
- **GOV-01.7** Approvers must be able to approve, reject, or return with comments; comments must be visible to the submitter.
- **GOV-01.8** Notifications for assignments, decisions, reminders and escalations must be delivered in-app, by email and through Microsoft Teams.
- **GOV-01.9** Publication must make the item visible in search and answers for its permitted audience and, where the item is managed in SharePoint, set the corresponding SharePoint approval or version state.
- **GOV-01.10** All workflow actions must be recorded in the audit trail with user, timestamp, action, comments and previous and new state.

Acceptance criteria

1. An administrator configures a three-stage workflow with parallel technical and compliance approval for a Policy asset type; a test policy moves through all stages and becomes searchable only after final approval.
2. An author cannot approve their own submission.
3. A stage past its due date escalates to the configured escalation contact.

GOV-02 Decommission workflows

Priority	Mandatory
RFP reference	2.1 Governance and Content Lifecycle Management, item 2

Description. Outdated knowledge that remains easy to find is a real operational risk. Decommission workflows let content be archived, retired, superseded by a newer version, or deleted, each through its own approval path. Superseded items point to their replacements, and AI answers cite the replacement instead. Deletion respects NYPA records retention and legal holds. A tamper-evident audit trail is kept even after an item is deleted.

Detailed requirements

- **GOV-02.1** The Platform must provide configurable decommission workflows for Archive, Retire, Supersede and Delete actions, each with its own approval stages and approvers.

- **GOV-02.2** Archive must remove the item from default search, recommendations and answers while keeping it accessible through an explicit “include archived” option.
- **GOV-02.3** Retire must mark the item as no longer valid, remove it from default search, recommendations and answers, and display a retired banner wherever it is accessed.
- **GOV-02.4** Supersede must link the item to its replacement; the superseded item must display a banner pointing to the replacement, and answers must cite the replacement instead.
- **GOV-02.5** Delete must remove the item from the index and, when configured, from the source system, subject to NYPA records retention rules and legal holds, which must block deletion.
- **GOV-02.6** Every decommission action must be recorded in a tamper-evident audit trail retained after deletion, including the item identity, title, actor, approvers, reason, timestamps and the replacement item for supersede.
- **GOV-02.7** Owners and watchers of an item must be notified when it is archived, retired, superseded or deleted.
- **GOV-02.8** Decommission actions must be available in bulk for a selected set of items, with a single approval.

Acceptance criteria

1. Superseding a procedure causes subsequent answers to cite the new version and the old version to display a banner linking to it.
2. Deletion of an item under legal hold is blocked with an explanatory message.
3. The audit trail for a deleted item remains available to Auditors.

GOV-03 Configurable ownership and stewardship

Priority	Mandatory
RFP reference	2.1 Governance and Content Lifecycle Management, item 3

Description. Every piece of governed content needs someone accountable for its accuracy. The Platform assigns an owner, and optionally stewards, to each item. Ownership is set directly or inherited from location, topic or asset type. When an owner leaves NYPA, their content is detected as orphaned and routed for reassignment. Ownership transfers are accepted by the recipient and audited, and each owner has a view of everything they are responsible for.

Detailed requirements

- **GOV-03.1** Every governed item must have an Owner and may have one or more Stewards, which can be individuals or groups.
- **GOV-03.2** Ownership must be assignable at item, folder, library, site, topic and asset-type level, with inheritance and item-level override.
- **GOV-03.3** Administrators must be able to define default ownership rules (for example, the author’s manager or a department’s knowledge lead).
- **GOV-03.4** The Platform must detect orphaned content when an owner leaves NYPA or is disabled in the directory, flag it on the content health dashboard and route reassignment tasks to the configured fallback owner.

- **GOV-03.5** Owners and stewards must be able to transfer ownership with acceptance by the recipient, recorded in the audit trail.
- **GOV-03.6** Bulk reassignment must be available to administrators and department approvers within their scope.
- **GOV-03.7** Each user must have a “My content” view showing the items they own or steward with lifecycle status, review dates and open tasks.

Acceptance criteria

1. Disabling a test owner in the directory results in their content being flagged as orphaned and reassignment tasks created within 24 hours.

GOV-04 Configurable review and validation workflows

Priority	Mandatory
RFP reference	2.1 Governance and Content Lifecycle Management, item 4

Description. Published knowledge must be confirmed periodically to remain accurate. Review intervals are set by asset type, department or item, and review tasks are generated automatically when content becomes due. Reviewers record an outcome: validated, updated, needs subject matter expert input, or recommend decommission. The “last validated” date is shown to users and used as a trust signal in search and answers. Any user can flag content as outdated, which creates a task for the owner.

Detailed requirements

- **GOV-04.1** The Platform must support periodic review workflows in which an owner or designated validator confirms that published content remains accurate, updates it, or initiates decommission.
- **GOV-04.2** Review intervals must be configurable per asset type, per department and per item (for example, 12 months for procedures, 24 months for reference material), and next review dates must be calculated automatically on publication and on each completed review.
- **GOV-04.3** Review workflows must be configurable in the same manner as publishing workflows (stages, approvers, due dates, escalation).
- **GOV-04.4** Reviewers must be able to record an outcome: Validated without change, Updated, Needs subject matter expert input, or Recommend decommission.
- **GOV-04.5** The Platform must display a “last validated” date and validator on the item preview and should use validation recency as a ranking and answer authority signal.
- **GOV-04.6** Ad hoc validation must be triggerable by users through a “report an issue” or “this is outdated” action on any item, creating a task for the owner.

Acceptance criteria

1. A procedure reaching its review date generates a review task; validation updates the next review date and the “last validated” display.
2. A user’s “this is outdated” report creates a task for the owner.

GOV-05 Stale content detection and automated review triggers

Priority	Mandatory
RFP reference	2.1 Governance and Content Lifecycle Management, item 5

Description. Review schedules alone do not catch all outdated content. Stale content detection applies configurable rules: review date passed, no edits or views over a period, links to retired content, broken links, departed owners and negative feedback. Each rule triggers an action such as notifying the owner, starting a review, lowering the item's search ranking or displaying a warning banner. This keeps the knowledge base current without relying on owners to remember.

Detailed requirements

- **GOV-05.1** The Platform must identify stale content using configurable rules combining: review date passed, no modification within a period, no views within a period, references to retired or superseded content, broken links, owner departed, and negative user feedback.
- **GOV-05.2** When an item meets a stale rule, the Platform must automatically trigger the configured review workflow and notify the owner.
- **GOV-05.3** Administrators must be able to configure the action per rule: notify only, trigger review, reduce ranking, or flag with a visible banner.
- **GOV-05.4** The Platform must detect broken links within indexed content (links to missing documents or pages) and report them to owners.
- **GOV-05.5** Stale rules must run on a schedule at least daily and on demand.

Acceptance criteria

1. An item not modified or viewed within the configured period triggers a review task and appears on the content health dashboard.
2. A document containing a link to a deleted page is reported as having a broken link.

GOV-06 Duplicate and near-duplicate detection

Priority	Mandatory
RFP reference	2.1 Governance and Content Lifecycle Management, item 6

Description. Copies of the same document across repositories cause confusion about which version is authoritative and clutter search results. The Platform detects exact duplicates by content regardless of name or location, detects near-duplicates such as format conversions and minor edits, and distinguishes them from related content. Duplicates are grouped in search results under the authoritative item. Contributors are warned before uploading a duplicate, and stewards can resolve duplicate clusters in bulk.

Detailed requirements

- **GOV-06.1** The Platform must detect exact duplicates across all connected sources using content hashing, independent of file name or location.

- **GOV-06.2** The Platform must detect near-duplicates (for example, minor edits, different formats of the same content, copies with changed headers) using content similarity with a configurable threshold.
- **GOV-06.3** The Platform must detect related documents (substantially overlapping topic and content) and present them as related rather than duplicate.
- **GOV-06.4** Duplicates must be grouped in search results so that one representative appears with an option to view the others; the representative must be the authoritative, most recent published item.
- **GOV-06.5** The Platform must warn contributors at upload when a duplicate or near-duplicate exists (ING-01.8).
- **GOV-06.6** Stewards must be able to review duplicate clusters, designate the authoritative item, and launch supersede or archive workflows for the others in bulk.
- **GOV-06.7** Duplicate statistics must be reported on the content health dashboard.

Acceptance criteria

1. The same document stored in SharePoint and Content Server is detected as an exact duplicate and grouped in results.
2. A PDF export of a Word document is detected as a near duplicate.
3. On a labeled sample, near-duplicate detection achieves at least 90 percent precision at the configured threshold.

GOV-07 Audit trail

Priority	Mandatory
RFP reference	2.1 Governance item 2 (“delete with audit trail”); Section 6 Security Guidelines

Description. A complete, tamper-evident audit trail supports accountability, compliance and investigation. The Platform records every significant action: content changes, metadata and tag decisions, workflow steps, ownership changes, decommission actions and configuration changes. Each record captures who acted, when, and what changed. Audit records cannot be altered, even by administrators. They are retained for a configurable period and can be searched, exported and streamed to NYPA’s security monitoring platform.

Detailed requirements

- **GOV-07.1** The Platform must record an audit event for every create, update, metadata change, tag acceptance or rejection, workflow action, ownership change, decommission action, permission-relevant configuration change, and administrative configuration change.
- **GOV-07.2** Audit events must include actor, timestamp (UTC), action, target, previous and new values where applicable, and client context.
- **GOV-07.3** Audit records must be immutable to all users, including administrators, and protected against tampering (for example, by write-once storage or hash chaining).
- **GOV-07.4** Auditors must be able to search, filter and export audit records.

- **GOV-07.5** Audit records must be retained for a configurable period, with a default of seven years, and be exportable to NYPA's security information and event management (SIEM) platform.

2.12 Analytics and dashboards

ANL-01 Content health dashboard

Priority	Mandatory
RFP reference	2.1 Analytics and Dashboards, item 1

Description. The content health dashboard tells knowledge managers how well the knowledge base is being maintained. It reports content recency, ownership coverage, review status, broken links, duplicates, metadata completeness and lifecycle state, with trends over time and filters by source, department, topic and owner. Every figure drills down to the underlying items, and users can act on them from the dashboard by assigning owners, starting reviews or launching decommission.

Detailed requirements

- **ANL-01.1** The Platform must provide a content health dashboard reporting, at minimum: content recency (age distribution and items not updated within configured periods), ownership coverage (share of items with an active owner, orphaned items), review status (current, due within 30 days, overdue, never reviewed), broken links, duplicates and near-duplicates, metadata completeness, and lifecycle state distribution.
- **ANL-01.2** Every metric must be filterable by source, business unit, department, site, asset type, topic and owner, and must support drill-down to the list of underlying items with the ability to act (assign owner, trigger review, launch decommission).
- **ANL-01.3** Dashboard visibility must be scoped by role: department approvers see their departments; administrators see the organization.
- **ANL-01.4** The dashboard must show trends over time for each metric.

ANL-02 Usage and adoption dashboard

Priority	Mandatory
RFP reference	2.1 Analytics and Dashboards, item 2

Description. The usage and adoption dashboard shows whether employees are using the Platform and getting value from it. It reports searches, questions, active users, click-through rates, time-to-answer, top queries and content, bounce rate, answer feedback and contribution activity, by department and location. Metrics follow consistent published definitions so that trends can be trusted. Individual identities are protected by default.

Detailed requirements

- **ANL-02.1** The Platform must report: number of searches and questions, unique and active users (daily, weekly, monthly), click-through rate, time-to-answer, top queries, top content, bounce rate, answer feedback ratings, contributions and approvals, and adoption by department and location.

- **ANL-02.2** Metrics must use the definitions below so that they are consistent and auditable.
- **ANL-02.3** Administrators must be able to compare periods and view trends.
- **ANL-02.4** Analytics must be presented without exposing individual user identities by default; person-level reporting must be restricted to authorized roles and subject to NYPA privacy policy.

Metric definitions

Metric	Definition
Search sessions	A sequence of search and answer interactions by one user with no gap longer than 30 minutes.
Click-through rate	Searches with at least one result click divided by all searches, for a period.
Click-through at k	Searches whose first click is at position k or higher, divided by all searches.
Mean reciprocal rank	Average over searches of 1 divided by the position of the first clicked result (0 if none).
Time-to-answer	Time from the first query of a session to the first long click (dwell of 30 seconds or more) or the first positively rated answer.
Bounce rate	Result clicks followed by a return to results within 10 seconds, divided by all result clicks.
Dwell time	Time between a result click and the user's next action in the Platform, capped at 30 minutes.
No-results rate	Searches returning zero results divided by all searches.
Reformulation rate	Searches followed within 60 seconds by a modified query, divided by all searches.
Abandonment rate	Searches with no click, no answer interaction and no follow-up, divided by all searches.
Answer rate	Questions for which an answer was generated divided by all questions.
Answer helpfulness	Answers rated helpful divided by answers rated.
Citation click rate	Answers with at least one citation opened divided by all answers.
Ownership coverage	Published items with an active owner divided by all published items.
Review compliance	Reviews completed on or before the due date divided by all reviews due in the period.
Metadata completeness	Published items meeting their asset type's completeness threshold divided by all published items.

Metric	Definition
Monthly active users	Distinct users with at least one search, answer, view or contribution in the calendar month.

ANL-03 Search quality metrics

Priority	Mandatory
RFP reference	2.1 Analytics and Dashboards, item 3

Description. Search quality metrics show where search is failing users and whether changes improve it. The Platform reports click-based measures of precision and recall, queries returning no or few results, query reformulation and abandonment, and dwell time. It also reports periodic offline relevance scores against a judged query set. Administrators can act on a failing query directly, for example by adding a synonym or a promoted result. Equivalent measures are reported for AI answer quality.

Detailed requirements

- **ANL-03.1** The Platform must report precision and recall proxies, including click-through at positions 1, 3 and 10, mean reciprocal rank of first click, query reformulation rate, and abandonment rate.
- **ANL-03.2** The Platform must report “no results” queries and low-result queries with frequency, and allow an administrator to act on them directly (add synonym, create promoted result, flag content gap).
- **ANL-03.3** The Platform must report dwell time on opened results, distinguishing short clicks from long clicks.
- **ANL-03.4** The Platform must support periodic offline relevance evaluation against a maintained judged query set and report nDCG@10 over time, including before and after relevance configuration changes.
- **ANL-03.5** The Platform must report answer quality signals: answer rate, decline rate, helpfulness rating, citation click rate, and confidence distribution.

ANL-04 Governance metrics

Priority	Mandatory
RFP reference	2.1 Analytics and Dashboards, item 4

Description. Governance metrics show whether lifecycle processes are being followed. The Platform reports review and validation compliance, workflow throughput and cycle times, rejection and escalation rates, acceptance rates for AI-generated metadata, and decommission activity. These metrics let NYPA hold departments and owners accountable and spot where processes are getting stuck.

Detailed requirements

- **ANL-04.1** The Platform must report review and validation compliance: reviews due, completed on time, completed late and overdue, by department, owner and asset type.
- **ANL-04.2** The Platform must report workflow performance: items in each stage, average and 90th percentile cycle time per stage, rejection and return rates, and escalations.

- **ANL-04.3** The Platform must report AI-assist governance: acceptance, edit and rejection rates of generated descriptions and tags, by field and over time.
- **ANL-04.4** The Platform must report decommission activity by action type and department.

ANL-05 Reporting export and BI integration

Priority	Mandatory
RFP reference	2.1 Analytics and Dashboards; Section 6 Architecture Guideline 5

Description. NYPA standardizes on Power BI and Tableau for enterprise reporting. The Platform exposes its analytics through a documented data feed and API that these tools can consume, and supports CSV and Excel export and scheduled email reports. This lets knowledge management metrics sit alongside other NYPA business intelligence rather than being confined to the Platform.

Detailed requirements

- **ANL-05.1** All dashboards must support export of the displayed data to CSV and Excel.
- **ANL-05.2** The Platform must expose the analytics data model through an authenticated API or data feed consumable by Microsoft Power BI and Tableau, and should provide a Power BI template.
- **ANL-05.3** Administrators must be able to schedule dashboard reports to be emailed to subscribers.
- **ANL-05.4** The analytics data model must be documented (see DES-06).

2.13 Integration and authoring

INT-01 Microsoft 365 as the single source of truth for storage

Priority	Mandatory
RFP reference	2.1 Integration and Authoring

Description. Microsoft 365 is NYPA's single source of truth for document storage, and the Platform must reinforce that rather than create a parallel repository. Content uploaded or created in the Platform is stored in NYPA-designated SharePoint locations. Confirmed metadata is written back to SharePoint columns. Edits made in Office are synchronized into the Platform, and version history remains in SharePoint. Platform-generated derivatives such as transcripts and embeddings are treated as protected index data, not as separate records.

Detailed requirements

- **INT-01.1** Content uploaded or created in the Platform must be stored in a NYPA-designated SharePoint Online library or OneDrive location; the Platform must not become a separate system of record for document files.
- **INT-01.2** Metadata confirmed in the Platform (summary, classification, taxonomy tags, owner, lifecycle state, review dates) must be written back to the corresponding SharePoint columns where a mapping is configured, so that SharePoint reflects the governed state.

- **INT-01.3** Edits made in SharePoint, including via Office desktop and Office for the web, must be synchronized into the Platform through the connector; the Platform must not overwrite newer SharePoint changes.
- **INT-01.4** Users must be able to open any Microsoft 365 document from the Platform for editing in Office for the web or the desktop application, subject to their SharePoint permissions.
- **INT-01.5** Version history must remain in SharePoint; the Platform must display and link to SharePoint versions and use them for “latest version” and “what changed” behavior.
- **INT-01.6** Platform-generated derivatives (extracted text, transcripts, thumbnails, embeddings) must be treated as index data, protected with the same permissions as the source, and regenerated from the source when needed.
- **INT-01.7** The Platform should integrate with Microsoft Teams for notifications, task actions (approve, reject, validate) and a search and ask experience.
- **INT-01.8** The Platform should respect Microsoft Purview sensitivity labels and retention labels applied in Microsoft 365.

Acceptance criteria

1. A document uploaded through the Platform is stored in the designated SharePoint library with confirmed metadata written to its columns.
2. An edit made in Word for the web appears in the Platform after the next incremental synchronization.

INT-02 Platform APIs

Priority	Mandatory
RFP reference	2.1 Integration and Authoring; 2.2 Knowledge Graph (consumable downstream); Section 6 Security Guidelines 1 and 2

Description. The Platform’s capabilities must be available to other NYPA systems and future AI applications, not only through its own user interface. Documented, versioned REST APIs cover search, answers, metadata, taxonomy, the knowledge graph, workflow tasks and analytics, secured with OAuth 2.0 and enforcing user permissions. Webhooks notify NYPA systems such as Azure DevOps or ServiceNow when content or workflow events occur.

Detailed requirements

- **INT-02.1** The Platform must expose documented REST APIs for search, answers, content metadata, taxonomy, knowledge graph, workflow tasks and analytics, protected by OAuth 2.0.
- **INT-02.2** API calls made on behalf of a user must enforce that user’s permissions; application-only access must be limited to explicitly granted scopes.
- **INT-02.3** APIs must be versioned, rate limited and described in an OpenAPI specification.
- **INT-02.4** The Platform should support webhooks or event notifications for content and workflow events so that NYPA systems (for example, Azure DevOps or ServiceNow) can react to them.

2.14 Administration and configuration

ADM-01 Administrative console

Priority	Mandatory
RFP reference	Section 6 Architecture Guideline 1

Description. NYPA must be able to configure and operate the Platform itself, without depending on the vendor for routine changes. A web-based administrative console provides access to every configurable capability in this specification. Administrative rights are divided into permission sets that can be delegated by function and scope, and every configuration change is audited.

Detailed requirements

- **ADM-01.1** All configurable aspects of the Platform described in this document must be manageable through a NYPA-accessible web administrative console, without vendor intervention.
- **ADM-01.2** Administrative capabilities must be divided into delegable permission sets (for example, connector administration, search administration, taxonomy, workflow, analytics, user and role administration) assignable to roles and scopes.
- **ADM-01.3** All configuration changes must be recorded in the audit trail with before and after values.

ADM-02 Configuration management, testing and promotion

Priority	Mandatory
RFP reference	Section 6 Architecture Guideline 2; Section 3.3 (configuration documentation and portability)

Description. NYPA must be able to test and verify configuration changes, upgrades and patches before they reach Production. Configuration is versioned, can be compared and rolled back, and can be exported as structured files for promotion between environments, storage in source control and business continuity recovery. Vendor upgrades arrive in Dev/Test/QA first with advance release notes. Feature flags allow new capabilities to be piloted with selected groups.

Detailed requirements

- **ADM-02.1** Configuration must be versioned, with the ability to compare versions and roll back.
- **ADM-02.2** Configuration must be exportable and importable as structured files (for example, JSON or YAML) so that configuration can be promoted from Dev/Test/QA to Production, stored in source control, and used to reconstitute the Platform for business continuity.
- **ADM-02.3** Vendor version upgrades and patches must be applied to the Dev/Test/QA environment first, with release notes provided in advance, and NYPA must be able to verify them before they reach Production within an agreed window.

- **ADM-02.4** The Platform should support feature flags so that new functionality can be enabled for pilot groups before general release.

ADM-03 User, role and group management

Priority	Mandatory
RFP reference	2.1 (role-based approvers, role-based discovery); Section 6 Security Guideline 2

Description. User accounts and access must follow NYPA's existing identity processes for joiners, movers and leavers. Accounts are provisioned and deprovisioned automatically from the NYPA directory. Platform roles are assigned to directory groups or individuals within a defined scope. Administrators can see the effective roles of any user at any time.

Detailed requirements

- **ADM-03.1** User accounts must be provisioned automatically from the NYPA directory using SCIM 2.0 or equivalent synchronization, including deprovisioning within one hour of directory disablement.
- **ADM-03.2** Platform roles must be assignable to directory groups and to individuals, with scope.
- **ADM-03.3** Administrators must be able to view the effective roles and scopes of any user.

3 Non-Functional Requirements

This section specifies the qualities the Platform must have in NYPA's environment: performance, availability, security, deployment, usability, operability, data management and responsible use of AI. It includes every Application Architecture and Security Architecture guideline in Section 6 of Attachment 1. Each requirement gives its description, its specific requirements and, where applicable, measurable targets. Performance and volume targets are proposed baselines to be confirmed with NYPA once actual content volumes and user populations are known.

3.1 Performance and scalability

NFR-PERF-01 Search responsiveness

Priority	Mandatory
Source	2.1 Unified Access, Search and Navigation; Section 3.4 performance testing

Description. Search is the most frequent interaction in the Platform, and users abandon search tools that feel slow. Search results, including facets and permission trimming, must return quickly enough to feel immediate, even at peak load and across the full multi-repository index.

Requirements

- **NFR-PERF-01.1** Server-side response time for a query returning the first page of results with facets must be 1.5 seconds or less at the 95th percentile and 3 seconds or less at the 99th percentile, at peak load.
- **NFR-PERF-01.2** Permission trimming must be included within these response times and must not be relaxed to meet them.
- **NFR-PERF-01.3** Type-ahead suggestions must be returned within 300 milliseconds at the 95th percentile.

NFR-PERF-02 Answer responsiveness

Priority	Mandatory
Source	2.1 Conversational Query and Answering

Description. Conversational answers involve retrieval, ranking and language model generation, which take longer than search. Streaming the answer as it is generated keeps the experience responsive, and limits on total generation time keep it usable.

Requirements

- **NFR-PERF-02.1** The first words of an answer must appear within 3 seconds at the 95th percentile.
- **NFR-PERF-02.2** A complete answer must be delivered within 15 seconds at the 95th percentile.

- **NFR-PERF-02.3** If answer generation is slow or unavailable, search results must still be displayed without delay.

NFR-PERF-03 User interface responsiveness

Priority	Mandatory
Source	2.1 Unified Access; Analytics and Dashboards

Description. Pages and dashboards must load promptly so that the Platform feels like a dependable daily tool, including for field staff on mobile devices.

Requirements

- **NFR-PERF-03.1** The landing page must be interactive within 2.5 seconds on a NYPA corporate network connection.
- **NFR-PERF-03.2** Document previews must begin rendering within 2 seconds for files up to 20 MB.
- **NFR-PERF-03.3** Dashboards must render within 5 seconds for organization-wide scope.

NFR-PERF-04 Index freshness

Priority	Mandatory
Source	2.1 item 2 (“respects original permissions”); 2.4 permission-aware indexing

Description. The index must reflect the current state of each source system. New and changed content should become findable promptly. Above all, revoked permissions and deleted content must stop appearing quickly, because a delay there is a disclosure risk.

Requirements

- **NFR-PERF-04.1** New and changed content and metadata must be searchable within 15 minutes of the change in the source system.
- **NFR-PERF-04.2** Permission revocations must take effect within 15 minutes for SharePoint Online and within 60 minutes for other sources. Where NYPA requires faster enforcement for designated sources, real-time permission checks must be used (CONN-02.5).
- **NFR-PERF-04.3** Deleted items must be removed from search, answers and recommendations within 15 minutes.

NFR-PERF-05 Processing throughput

Priority	Mandatory
Source	2.1 Automated Metadata Generation; 2.2 Auto-Classification and Tagging

Description. AI enrichment must keep pace with contribution and with large batch jobs such as initial migration or reclassification after a taxonomy change. Contributors should receive suggestions while they are still on the upload screen.

Requirements

- **NFR-PERF-05.1** Summary and tag suggestions for a 50-page document must be available within 2 minutes of upload completion.
- **NFR-PERF-05.2** Batch classification must process at least 500 documents in 30 minutes, and throughput must scale roughly linearly with added capacity.
- **NFR-PERF-05.3** Video and audio transcription must complete within the duration of the recording for files up to 2 hours.

NFR-PERF-06 Concurrency and scale

Priority	Mandatory
Source	Section 3.4 performance testing

Description. The Platform must serve the full NYPA and Canals workforce and the full indexed corpus, and have room to grow without architectural change.

Requirements

- **NFR-PERF-06.1** The Platform must support at least 1,000 concurrent users and a sustained 50 queries per second while meeting NFR-PERF-01.
- **NFR-PERF-06.2** The Platform must support at least 5 million indexed items and 50 million passages, with headroom to 10 million items without architectural change.
- **NFR-PERF-06.3** Capacity must be scalable horizontally without service interruption.

3.2 Availability and disaster recovery

NFR-AVL-01 Service availability

Priority	Mandatory
Source	Section 6 Architecture Guideline 8

Description. Once the Platform is the primary way employees find knowledge, an outage directly affects work, including in operational settings. The Platform must be available whenever NYPA staff need it, with maintenance planned around NYPA's business.

Requirements

- **NFR-AVL-01.1** Production availability must be at least 99.9 percent per month, excluding agreed maintenance windows.
- **NFR-AVL-01.2** Planned maintenance must be announced at least 5 business days in advance and scheduled outside NYPA business hours.
- **NFR-AVL-01.3** Availability must be reported to NYPA monthly.

NFR-AVL-02 High availability

Priority	Mandatory
Source	Section 6 Architecture Guideline 8

Description. No failure of a single component should take the Platform offline. All tiers must be redundant, and the Platform must degrade gracefully when a dependency is impaired.

Requirements

- **NFR-AVL-02.1** All Platform services must run redundantly across at least two availability zones, with no single point of failure.
- **NFR-AVL-02.2** If the answer service is unavailable, search must remain fully functional.
- **NFR-AVL-02.3** If a connector or source system is unavailable, previously indexed content must remain searchable, and its permissions must remain enforced.

NFR-AVL-03 Disaster recovery

Priority	Mandatory
Source	Section 6 Architecture Guideline 8 (multi-site disaster recovery)

Description. The Platform must recover from the loss of an entire hosting region within defined limits on data loss and downtime, and recovery must be proven by regular testing.

Requirements

- **NFR-AVL-03.1** The Platform must be recoverable in a secondary United States region.
- **NFR-AVL-03.2** The recovery point objective must be 1 hour or less for configuration, governance data, conversation history and audit records. Index data may be rebuilt from source systems.
- **NFR-AVL-03.3** The recovery time objective must be 4 hours or less.
- **NFR-AVL-03.4** Daily encrypted backups must be retained for at least 35 days.
- **NFR-AVL-03.5** Restore and failover must be tested at least twice a year, with results provided to NYPA.

3.3 Security

NFR-SEC-01 Data extraction and data model documentation

Priority	Mandatory
Source	Section 6 Security Guideline 1

Description. NYPA must remain in control of its data and must never be locked in. All data held by the Platform must be extractable in documented formats, and NYPA must have documentation of the data model.

Requirements

- **NFR-SEC-01.1** All NYPA data, including content, metadata, tags, taxonomy, knowledge graph, workflow history, audit records, conversation history and analytics, must be extractable through APIs or bulk export in documented, non-proprietary formats.
- **NFR-SEC-01.2** Complete data model documentation (entities, attributes, relationships) must be provided to NYPA and kept current with each release.

NFR-SEC-02 Workforce authentication

Priority	Mandatory
Source	Section 6 Security Guideline 2

Description. NYPA employees must sign in with their existing NYPA identity, and NYPA's identity controls must apply. These include multi-factor authentication and conditional access.

Requirements

- **NFR-SEC-02.1** User authentication must integrate with NYPA Active Directory through Microsoft Entra ID using SAML 2.0. OpenID Connect must also be supported.
- **NFR-SEC-02.2** NYPA multi-factor authentication and conditional access policies must apply to Platform sign-in.
- **NFR-SEC-02.3** The Platform must not maintain separate local passwords for NYPA users.
- **NFR-SEC-02.4** Session timeout and re-authentication intervals must be configurable by NYPA.

NFR-SEC-03 API authentication

Priority	Mandatory
Source	Section 6 Security Guideline 2

Description. Systems that call the Platform's APIs must authenticate with modern, token-based methods that can be scoped and revoked.

Requirements

- **NFR-SEC-03.1** APIs must use OAuth 2.0, with client credentials for application access and on-behalf-of flows for user-delegated access, or an equivalent token-based mechanism.
- **NFR-SEC-03.2** Long-lived static API keys must not be required.
- **NFR-SEC-03.3** Tokens must be scoped to the minimum permissions required.

NFR-SEC-04 External user federation

Priority	Mandatory
Source	Section 6 Security Guideline 3

Description. Where contractors or partners need access, they must authenticate through their own identity provider under NYPA's control, and receive only restricted access.

Requirements

- **NFR-SEC-04.1** External, non-NYPA users must authenticate through federation with a third-party identity provider, for example Entra ID B2B or another SAML 2.0 or OpenID Connect provider.
- **NFR-SEC-04.2** External users must be restricted to the Guest role unless NYPA explicitly grants more.

NFR-SEC-05 Tenancy and isolation

Priority	Mandatory
Source	Section 6 Security Guideline 4

Description. NYPA prefers single-tenant solutions. The Platform must be deployable as an instance dedicated to NYPA. Any shared components must meet NIST SP 800-53 controls.

Requirements

- **NFR-SEC-05.1** The Platform must be deployable as a single-tenant instance with dedicated compute, storage, search index and encryption keys.
- **NFR-SEC-05.2** Any shared component must be documented, and its security controls must meet the NIST SP 800-53 moderate baseline.

NFR-SEC-06 Encryption

Priority	Mandatory
Source	Section 6 Security Guideline 5

Description. All NYPA data must be protected by strong encryption wherever it is stored or transmitted, using algorithms and key lengths consistent with current NIST guidance.

Requirements

- **NFR-SEC-06.1** Data at rest, including indexes, embeddings, transcripts, backups and logs, must be encrypted with AES-256 or stronger.
- **NFR-SEC-06.2** NYPA should be able to supply and control its own encryption keys (customer-managed keys).
- **NFR-SEC-06.3** Data in transit must be encrypted end to end, including between internal services.

NFR-SEC-07 Independent security assurance

Priority	Mandatory
Source	Section 6 Security Guideline 6

Description. Because Platform data is stored outside NYPA's private cloud, NYPA needs independent evidence that the hosting service is operated securely.

Requirements

- **NFR-SEC-07.1** The vendor must provide a current SOC 2 Type II report for the hosting service. Where a SOC report is not applicable, the vendor must provide a third-party penetration test report completed within the previous 12 months.
- **NFR-SEC-07.2** Assurance reports must be renewed and provided annually.

NFR-SEC-08 Non-cacheable sensitive screens

Priority	Mandatory
Source	Section 6 Security Guideline 7

Description. Confidential content must not be left behind in browser or proxy caches, where it could be retrieved by someone else using the same device or network path.

Requirements

- **NFR-SEC-08.1** All screens and API responses containing confidential or sensitive information, including search results, previews, thumbnails and answers, must be served with headers that prevent caching (Cache-Control: no-store).
- **NFR-SEC-08.2** Service workers and offline features must not store sensitive content unless NYPA approves.

NFR-SEC-09 Transport security

Priority	Mandatory
Source	Section 6 Security Guideline 8

Description. All network communication must use current, strong transport encryption, and older protocols must be disabled.

Requirements

- **NFR-SEC-09.1** All HTTP and other TLS-protected communication must use TLS 1.2 or later with key sizes of 128 bits or larger.
- **NFR-SEC-09.2** SSL, TLS 1.0, TLS 1.1 and weak cipher suites must be disabled.

NFR-SEC-10 Data residency

Priority	Mandatory
Source	Section 6 Security Guideline 9

Description. NYPA data must stay within the United States. This includes data processed by AI models, and access by support personnel.

Requirements

- **NFR-SEC-10.1** All NYPA data, including content sent to AI models, must be stored and processed only within the continental United States or its territories, unless NYPA approves otherwise in writing.
- **NFR-SEC-10.2** Vendor personnel with access to NYPA data must be located in the United States.

NFR-SEC-11 Access control and least privilege

Priority	Mandatory
Source	NIST SP 800-53 Access Control family

Description. Access within the Platform, including administrative access, must follow least privilege and segregation of duties.

Requirements

- **NFR-SEC-11.1** The Platform must implement role-based access control with roles scoped to organization, business unit, department or topic.
- **NFR-SEC-11.2** Administrative duties must be separable, so that, for example, connector administrators cannot change audit settings.
- **NFR-SEC-11.3** Privileged actions must require re-authentication where NYPA policy requires it.

NFR-SEC-12 Security logging and monitoring

Priority	Mandatory
Source	NIST SP 800-53 Audit and Accountability family

Description. NYPA's security operations team must be able to see security-relevant activity in the Platform in near real time, alongside other enterprise systems.

Requirements

- **NFR-SEC-12.1** Sign-ins, failed sign-ins, privilege changes, configuration changes, data exports and anomalous access patterns must be logged.
- **NFR-SEC-12.2** Security logs must be streamable to NYPA's security information and event management (SIEM) platform in near real time.

NFR-SEC-13 Application and AI threat protection

Priority	Mandatory
Source	NIST SP 800-53 System and Information Integrity family

Description. The Platform must resist common web application attacks and attacks specific to AI applications, such as prompt injection.

Requirements

- **NFR-SEC-13.1** The Platform must be protected by a web application firewall and rate limiting.
- **NFR-SEC-13.2** The Platform must address the OWASP Top 10 and the OWASP Top 10 for Large Language Model Applications.
- **NFR-SEC-13.3** Uploaded files must be scanned for malware before processing.

NFR-SEC-14 Vulnerability and incident management

Priority	Mandatory
Source	NIST SP 800-53 Incident Response and Risk Assessment families

Description. Vulnerabilities must be fixed promptly, and NYPA must be told quickly about any incident affecting its data.

Requirements

- **NFR-SEC-14.1** Critical vulnerabilities must be remediated within 7 days and high vulnerabilities within 30 days.
- **NFR-SEC-14.2** NYPA must be notified of any confirmed security incident affecting its data within 24 hours.

NFR-SEC-15 Secrets and credential management

Priority	Mandatory
Source	NIST SP 800-53 Identification and Authentication family

Description. Connector credentials, certificates and keys give access to NYPA repositories and must be protected and rotated.

Requirements

- **NFR-SEC-15.1** Secrets, certificates and connector credentials must be stored in a managed secrets vault and never in configuration files or code.
- **NFR-SEC-15.2** Credentials must be rotated according to NYPA policy, without service interruption.

3.4 Deployment and environments

NFR-DEP-01 Hosting model

Priority	Mandatory
Source	Section 6 Architecture Guideline 4

Description. The Platform must be offered in a deployment model NYPA prefers, on major United States cloud infrastructure, with portability across cloud providers.

Requirements

- **NFR-DEP-01.1** The Platform must be available as Software as a Service hosted on AWS or Azure in United States regions.
- **NFR-DEP-01.2** The Platform should be deployable on at least two major cloud providers.
- **NFR-DEP-01.3** Deployment as Platform as a Service on Azure, or within NYPA's own Azure or AWS subscription (Infrastructure as a Service), must be supported as an option.

NFR-DEP-02 Environments

Priority	Mandatory
Source	Section 3.3; Section 6 Architecture Guideline 2

Description. NYPA must be able to test configuration, upgrades and patches in a separate environment before they reach Production.

Requirements

- **NFR-DEP-02.1** Two environments must be provided: Dev/Test/QA and Production.
- **NFR-DEP-02.2** The environments must have equivalent configuration and software versions, and separate data.

- **NFR-DEP-02.3** Vendor upgrades and patches must be deployed to Dev/Test/QA before Production, with a verification window agreed with NYPA.

NFR-DEP-03 Hybrid connectivity

Priority	Mandatory
Source	Section 3.3 (hybrid multi-cloud environment)

Description. Some NYPA repositories, such as Content Server, may be on premises. The Platform must reach them securely without exposing them to the internet.

Requirements

- **NFR-DEP-03.1** The Platform must connect to on-premises repositories through a NYPA-approved mechanism, such as private link, site-to-site VPN or an outbound-only on-premises connector agent.
- **NFR-DEP-03.2** Connectivity must not require inbound firewall openings to NYPA networks from the internet.

NFR-DEP-04 Reproducibility and portability

Priority	Mandatory
Source	Section 3.3 (configuration portability for business continuity)

Description. Environments and configuration must be reproducible, so that the Platform can be rebuilt or reconfigured for business continuity and recovery.

Requirements

- **NFR-DEP-04.1** Infrastructure must be defined as code.
- **NFR-DEP-04.2** All Platform configuration must be exportable and importable as structured files (ADM-02).

3.5 Usability and accessibility

NFR-UX-01 Accessibility

Priority	Mandatory
Source	New York State accessibility policy

Description. The Platform must be usable by all NYPA employees, including those who use assistive technologies.

Requirements

- **NFR-UX-01.1** The user interface must conform to WCAG 2.1 Level AA.
- **NFR-UX-01.2** The vendor must provide an accessibility conformance report and remediate identified gaps.

- **NFR-UX-01.3** Video content must have searchable transcripts, which also serve as captions.

NFR-UX-02 Supported clients

Priority	Mandatory
Source	Section 6 Architecture Guideline 6

Description. The Platform must work on the devices and browsers NYPA uses, including mobile devices for field staff.

Requirements

- **NFR-UX-02.1** The Platform must support the latest two major versions of Microsoft Edge and Google Chrome on Windows 10 and Windows 11.
- **NFR-UX-02.2** The Platform must provide a responsive experience on iOS devices in Safari.
- **NFR-UX-02.3** The Platform should provide an iOS application or progressive web application for field users.

NFR-UX-03 Look and feel

Priority	Mandatory
Source	Section 6 Architecture Guideline 7

Description. The Platform must look like part of NYPA's digital estate and fit with NYPA's intranet experience.

Requirements

- **NFR-UX-03.1** The user interface must be customizable to NYPA's standard look and feel, including logo, colors and typography.
- **NFR-UX-03.2** The Platform should support embedding its search and answer components within a Sitecore-based intranet.
- **NFR-UX-03.3** User-facing labels must be configurable to use NYPA terminology.

NFR-UX-04 Ease of use

Priority	Mandatory
Source	Section 2 Solution Vision (knowledge in the flow of work)

Description. Adoption depends on employees finding the Platform easy to use without formal training.

Requirements

- **NFR-UX-04.1** A new user must be able to search, ask a question, and upload and submit a document without training. This must be verified with representative NYPA users during acceptance testing.
- **NFR-UX-04.2** Contextual help must be available on every page.

3.6 Operations and support

NFR-OPS-01 Monitoring and alerting

Priority	Mandatory
Source	Section 3.5 (post-production support)

Description. Problems must be detected and acted on before users notice them.

Requirements

- **NFR-OPS-01.1** All services, connectors and processing pipelines must have health monitoring, with alerts to NYPA and vendor support channels.
- **NFR-OPS-01.2** Application and pipeline logs must be retained for at least 90 days and be available to NYPA support staff.
- **NFR-OPS-01.3** The vendor must provide a service status page and an incident communication process.

NFR-OPS-02 Support

Priority	Mandatory
Source	Section 3.5 (post go-live support)

Description. NYPA needs dependable vendor support, with the fastest response for incidents that affect operations.

Requirements

- **NFR-OPS-02.1** Vendor support must be available during NYPA business hours at minimum.
- **NFR-OPS-02.2** Severity 1 incidents must be supported 24 hours a day, 7 days a week.
- **NFR-OPS-02.3** Response and resolution targets by severity must be defined in the service level agreement.

3.7 Data management and privacy

NFR-DAT-01 Retention and legal hold

Priority	Mandatory
Source	2.1 Governance (decommission with audit trail)

Description. NYPA must control how long each kind of data is kept, and must be able to preserve data when legally required.

Requirements

- **NFR-DAT-01.1** Retention periods for conversation history, analytics events, audit records and logs must be configurable by NYPA.
- **NFR-DAT-01.2** The Platform must support legal hold on content and audit records, which blocks deletion.
- **NFR-DAT-01.3** The Platform should respect Microsoft Purview retention labels applied in Microsoft 365.

NFR-DAT-02 Privacy

Priority	Mandatory
Source	NYPA privacy policy

Description. Personalization and analytics use information about employees' activity. This must be limited to what is necessary and protected.

Requirements

- **NFR-DAT-02.1** Personal data used for personalization and analytics must be limited to what is necessary for those purposes.
- **NFR-DAT-02.2** Analytics must be pseudonymized by default, with person-level reporting restricted to authorized roles.
- **NFR-DAT-02.3** Users must be able to view and delete their own conversation history.

NFR-DAT-03 Data return at exit

Priority	Mandatory
Source	Section 6 Security Guideline 1

Description. At the end of the contract, NYPA must get all of its data back, and the vendor must remove its copies.

Requirements

- **NFR-DAT-03.1** On termination, the vendor must return all NYPA data in documented formats within an agreed period.
- **NFR-DAT-03.2** The vendor must certify deletion of NYPA data, including from backups, within an agreed period.

3.8 Responsible AI

NFR-AI-01 Data protection in AI processing

Priority	Mandatory
Source	Section 6 Security Guideline 9; 2.1 Conversational Query and Answering

Description. NYPA content processed by AI models must remain protected, inside the United States, and must never be used to train models outside NYPA's control.

Requirements

- **NFR-AI-01.1** NYPA content, prompts and outputs must not be used to train or improve any model outside NYPA's dedicated instance.
- **NFR-AI-01.2** AI models must be hosted in United States regions, either within the Platform's own infrastructure or through an enterprise model service with zero data retention.
- **NFR-AI-01.3** Only content the requesting user is permitted to access may be included in model inputs.

NFR-AI-02 Human oversight

Priority	Mandatory
Source	2.1 Automated Metadata Generation (human-in-the-loop review)

Description. People, not models, remain accountable for authoritative metadata and vocabulary. NYPA must be able to control where AI is used.

Requirements

- **NFR-AI-02.1** All AI-generated metadata and taxonomy suggestions must be reviewable by a person before becoming authoritative.
- **NFR-AI-02.2** AI-generated content must be labeled as such wherever it is displayed.
- **NFR-AI-02.3** Administrators must be able to disable any AI capability independently, by organization, business unit or source.

NFR-AI-03 Transparency and traceability

Priority	Mandatory
Source	2.1 Conversational Query and Answering (confidence and explainability)

Description. Every AI output must be traceable to the model, instructions and inputs that produced it, so that it can be reviewed, explained and audited.

Requirements

- **NFR-AI-03.1** For every AI output, the Platform must record the model and version, the prompt template version, references to the inputs, and the output.
- **NFR-AI-03.2** Each AI capability must be documented with its purpose, model, known limitations and evaluation results.

NFR-AI-04 Quality evaluation and regression control

Priority	Mandatory
Source	2.1 Conversational Query and Answering; Automated Metadata Generation

Description. Changes to models, prompts or retrieval must never silently degrade answer or tagging quality.

Requirements

- **NFR-AI-04.1** The Platform must maintain NYPA-specific evaluation sets for answers, summaries and classification, developed with NYPA subject matter experts.
- **NFR-AI-04.2** Any change affecting AI outputs must be evaluated against these sets in Dev/Test/QA, and must not fall below agreed thresholds before promotion to Production.
- **NFR-AI-04.3** Answer quality must be monitored in Production through user feedback and periodic sampled review.

NFR-AI-05 AI safety and robustness

Priority	Mandatory
Source	NIST AI Risk Management Framework

Description. The answer service must resist manipulation and must not disclose protected information.

Requirements

- **NFR-AI-05.1** The Platform must defend against prompt injection in indexed content and in user input.
- **NFR-AI-05.2** The Platform must not reveal system instructions, configuration or content outside the user's permissions in response to adversarial questions.
- **NFR-AI-05.3** The Platform must redact sensitive data types that NYPA designates from answers when configured.
- **NFR-AI-05.4** Adversarial testing of the answer service must be performed before go-live and after significant changes.

4 Data, Integration and Analytics Design Requirements

Section 2.4 of Attachment 1 lists design artifacts that describe how the Enterprise Knowledge Platform's data, integrations and analytics work. The Platform and its delivery documentation must provide each of the following. Each artifact is delivered as part of the Consolidated Solution Design and kept current in the As-built Documentation.

DES-01 Unified search integration framework

The framework must document how secure, permission-aware indexing operates across Microsoft 365, Content Server, ServiceNow and intranet repositories. It must include:

- The connector inventory, with source system, API used, authentication method, content scope, metadata captured and synchronization method for each.
- The permission model mapping from each source system's access model to the Platform's normalized ACL model, including group resolution and the handling of deny entries, sharing links and inheritance.
- Synchronization schedules, change-detection mechanisms and latency targets.
- Failure handling, retry, throttling and monitoring.
- The process to add a new source through the connector framework.

DES-02 Semantic and hybrid search pipeline design

The design must describe how keyword retrieval and ontology-driven relevance combine. It must include:

- Query processing steps: language detection, spelling correction, tokenization, entity and concept recognition, taxonomy expansion, acronym handling and intent detection.
- Index design: fields, analyzers, vector embeddings (model, dimension, chunking strategy), security fields and facet fields.
- Retrieval and fusion: keyword retrieval, vector retrieval, fusion method and re-ranking.
- Ranking signals and their default weights: text relevance, semantic similarity, taxonomy match, graph proximity, freshness, authority, lifecycle state, validation recency, personalization and popularity.
- Tuning and evaluation method, including the judged query set and metrics.

DES-03 Knowledge graph integration approach

The design must show how entities, relationships and semantic signals improve search, navigation and conversational answers. It must include:

- The graph schema: node types, edge types, properties and identifiers, aligned to the enterprise taxonomy and ontology.
- Population sources and update triggers.
- How the graph is used in ranking (SEM-03.3), navigation (SEM-03.4) and answer retrieval (SEM-03.6), with worked examples using NYPA scenarios.

- Permission enforcement for graph traversal.
- Consumption patterns for downstream systems and AI applications: API, export formats and refresh frequency.

DES-04 Metadata and taxonomy/ontology mapping specification

The specification must ensure consistent tagging, faceted navigation and semantic enrichment across all connected systems. It must include:

- The Platform metadata schema: field name, type, cardinality, term set binding, mandatory status per asset type, and source.
- Field-by-field mappings from SharePoint columns and term store, Content Server categories and attributes, ServiceNow article fields and intranet page metadata to the Platform schema.
- The mapping between Platform term sets, the enterprise taxonomy tool and the SharePoint term store, including the master for each term set and synchronization direction.
- Rules for reconciling conflicting values from different sources.
- The facet configuration derived from the schema.

DES-05 Ingestion and enrichment workflow definitions

The definitions must describe the processing of each item from discovery to publication. They must include:

- Pipeline stages: discovery, retrieval, malware scan, format detection, text extraction, OCR, transcription, chunking, embedding, summary generation, classification and auto-tagging, entity extraction, duplicate detection, completeness check, human review routing, publication and write-back.
- For each stage: inputs, outputs, error handling, retry policy and observability.
- The human review steps: who reviews what, under which thresholds, within which service levels.
- Differences between uploaded content and connector-ingested content.

DES-06 Platform analytics data model

The model must support content health, adoption, search quality and governance insights. It must include:

- Event schema for user interactions (search issued, result shown, result clicked, dwell, answer generated, citation clicked, feedback given, item viewed, item followed).
- Fact and dimension tables for content, users (pseudonymized where required), queries, answers, workflows and reviews.
- Metric definitions consistent with those in ANL-02.
- Data retention and privacy controls.
- Access methods for Power BI and Tableau.

DES-07 Retrieval-augmented Q&A architecture

The architecture must show how data sources, citations and permission-filtered responses are generated and evaluated. It must include:

- The end-to-end request flow from question to streamed answer: query understanding, conversation context handling, retrieval with permission filtering, graph expansion, re-ranking, context assembly, prompt construction, generation, citation linking and confidence scoring.
- The language models used, where they are hosted, data residency, and the controls that prevent NYPA data from being used to train third-party models.
- Safeguards: grounding requirements, refusal behavior, prompt injection defenses, output filtering and sensitive data handling.
- The evaluation method: evaluation sets, metrics (faithfulness, citation precision, answer relevance, refusal accuracy, permission leakage), cadence and regression gates for model or prompt changes.

Appendix A. User Roles

The Platform uses role-based access control. Roles are assigned from NYPA directory groups, so that joiners, movers and leavers are handled through existing identity processes.

Role	Typical holders	Principal responsibilities in the Platform
Platform Administrator	EKM program team, NYPA IT application owners	Configure connectors, landing pages, search relevance, facets, templates, workflows, metadata rules, roles and dashboards. Access all audit logs.
Taxonomy Manager	EKM taxonomists	Maintain the managed term sets and synonyms, review AI-suggested terms, manage synchronization with the enterprise taxonomy tool.
Content Approver	Department knowledge leads, designated reviewers	Review and approve submissions in publishing and decommission workflows for their scope.
Content Owner	Subject matter experts accountable for specific assets	Accountable for accuracy; receives review and stale-content tasks; approves changes to owned content.
Content Steward	Knowledge coordinators	Maintains metadata quality, responds to feedback, prepares content for review.
Contributor	Any employee permitted to submit content	Uploads content, completes metadata, submits for review, edits own drafts.
Knowledge Consumer	All NYPA and Canals employees	Searches, asks questions, browses, follows topics, saves searches, gives feedback.
Analytics Viewer	Managers, EKM leadership	Views dashboards for their scope; cannot change content or configuration.
Auditor	Internal audit, compliance	Read-only access to audit trails, workflow histories and governance reports.
Guest	Contractors and external partners, where permitted	Restricted read-only access to explicitly shared content, authenticated through a federated identity provider.

A user may hold several roles. Roles may be scoped to the whole organization, to a business unit (for example, Canals), to a department, or to a topic area.

Appendix B. Requirements Traceability Matrix

This matrix maps each line of RFP Attachment 1, Section 2.1, together with the Section 2.4 Enterprise Knowledge Platform design items and the Section 6 guidelines, to the requirements in this document.

Section 2.1 Enterprise KM Platform Capabilities

RFP capability	RFP requirement	Requirements
Unified Access, Search and Navigation	Single knowledge platform with a landing page customized to NYPA and Canals context	ACC-01, ACC-02
Unified Access, Search and Navigation	Unified search across SharePoint (mandatory), Content Server (target), ServiceNow (target), respecting original permissions	GEN-01, SRCH-01, CONN-01, CONN-02, CONN-03, CONN-04, CONN-06
Unified Access, Search and Navigation	Semantic/hybrid search combining keyword and ontology-driven results	SRCH-02, SEM-01
Unified Access, Search and Navigation	Faceted navigation based on taxonomy/ontology	SRCH-03, SRCH-04
Personalization and Role-Aware Experience	Role-based discovery, personalized homepages, topic hubs, saved searches and watchlists	PERS-01, PERS-02, PERS-03, PERS-04, PERS-05
Personalization and Role-Aware Experience	Recommendations and trending content in users' domains	PERS-06, PERS-07
Conversational Query and Answering	Natural language questions with direct answers alongside search results	CQA-01
Conversational Query and Answering	Retrieval-augmented answers citing authoritative sources and respecting permissions	CQA-02, GEN-01, CONN-02, NFR-AI-01
Conversational Query and Answering	Follow-up questions and conversational refinement	CQA-03
Conversational Query and Answering	Confidence scores and explainability	CQA-04, NFR-AI-03
Knowledge Content Coverage	Multiple asset formats: documents, PPTs, PDFs, videos, images	COV-01
Knowledge Content Coverage	Site pages, wikis and knowledge articles within connected systems	COV-02, CONN-01, CONN-04, CONN-05
Knowledge Content Coverage	Indexing and previews of common file types, thumbnails for visual content	COV-03

RFP capability	RFP requirement	Requirements
Knowledge Content Coverage	Configurable templates for common knowledge asset types	COV-04
Content Enrichment and Semantic Layer Integration	Managed metadata fields with hierarchical terms and synonyms	SEM-01
Content Enrichment and Semantic Layer Integration	Integration with NYPA/Canals taxonomy/ontology tool	SEM-02, SEM-04
Content Enrichment and Semantic Layer Integration	Knowledge graph enriched search, navigation and conversational querying	SEM-03
Content Ingestion	Easy drag-and-drop upload	ING-01
Automated Metadata Generation	Brief description at ingest with human-in-the-loop review	META-01
Automated Metadata Generation	Auto-classification and auto-tagging at ingest and update with human-in-the-loop review	META-02
Automated Metadata Generation	Minimum metadata completeness thresholds before publishing	META-03
Governance and Content Lifecycle Management	Publishing workflows with configurable stages and role-based approvers	GOV-01
Governance and Content Lifecycle Management	Decommission workflows: archive, retire, supersede, delete with audit trail	GOV-02, GOV-07
Governance and Content Lifecycle Management	Configurable ownership and stewardship	GOV-03
Governance and Content Lifecycle Management	Configurable review and validation workflows	GOV-04
Governance and Content Lifecycle Management	Stale content detection and automated review triggers	GOV-05
Governance and Content Lifecycle Management	Duplicate and near-duplicate detection	GOV-06
Analytics and Dashboards	Content health dashboard	ANL-01
Analytics and Dashboards	Usage and adoption dashboard	ANL-02
Analytics and Dashboards	Search quality metrics	ANL-03
Analytics and Dashboards	Governance metrics	ANL-04, ANL-05
Integration and Authoring	Microsoft 365 as single source of truth storage	INT-01, INT-02

Section 2.4 Data, Integration and Analytics (Enterprise Knowledge Platform)

RFP design item	Requirements
Unified search integration framework with permission-aware indexing across M365, Content Server, ServiceNow and intranet	DES-01, CONN-01 to CONN-06
Semantic/hybrid search pipeline design	DES-02, SRCH-02
Knowledge graph integration approach	DES-03, SEM-03
Metadata and taxonomy/ontology mapping specification	DES-04, SEM-01, SEM-02
Ingestion and enrichment workflow definitions	DES-05, ING-01, META-01 to META-03
Platform analytics data model	DES-06, ANL-01 to ANL-05
Retrieval-augmented Q&A architecture	DES-07, CQA-01 to CQA-04, NFR-AI-01 to NFR-AI-05

Section 6 Application and Security Architecture Guidelines

Guideline	Requirements
Architecture 1: NYPA-accessible configuration mechanism	ADM-01
Architecture 2: Test and verify configuration, upgrades and patches before production	ADM-02, NFR-DEP-02
Architecture 3: Vendor-certified, configurable connectors	CONN-06.5, CONN-05.2
Architecture 4: SaaS, IaaS or PaaS deployment; multi-cloud preference	NFR-DEP-01, NFR-DEP-03
Architecture 5: Tableau or Power BI integration	ANL-05
Architecture 6: Windows 10, Chrome and Edge, iOS	NFR-UX-02
Architecture 7: Sitecore or NYPA look and feel	ACC-01.4, NFR-UX-03
Architecture 8: High availability and multi-site disaster recovery	NFR-AVL-01 to NFR-AVL-03
Security 1: Data extraction and data model documentation	NFR-SEC-01, NFR-DAT-03
Security 2: AD integration via SAML 2.0, OAuth for APIs	NFR-SEC-02, NFR-SEC-03, ADM-03
Security 3: Federation for external users	NFR-SEC-04
Security 4: Single tenancy; NIST 800-53 for multi-tenant	NFR-SEC-05
Security 5: Encryption at rest and in transit	NFR-SEC-06
Security 6: SOC report or penetration test	NFR-SEC-07
Security 7: Non-cacheable sensitive screens	NFR-SEC-08

Guideline	Requirements
Security 8: TLS 1.2 or later	NFR-SEC-09
Security 9: Data residency in the United States	NFR-SEC-10, NFR-AI-01.2